

Autotrophic growth of *E. coli* is achieved by a small number of genetic changes

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Abstract

Synthetic autotrophy is a promising avenue to sustainable bioproduction from CO₂. Here, we use iterative laboratory evolution to generate several distinct autotrophic strains. Utilising this genetic diversity, we identify that just three mutations are sufficient for *E. coli* to grow autotrophically, when introduced alongside non-native energy (formate dehydrogenase) and carbon-fixing (RuBisCO, phosphoribulokinase, carbonic anhydrase) modules. The mutated genes are involved in glycolysis (*pgi*), central-carbon regulation (*crp*), and RNA transcription (*rpoB*). The *pgi* mutation reduces the enzyme's activity, thereby stabilising the carbon-fixing cycle by capping a major branching flux. For the other two mutations, we observe down-regulation of several metabolic pathways and increased expression of native genes associated with the carbon-fixing module (*rpiB*) and the energy module (*fdoGH*), as well as an increased ratio of NADH/NAD⁺ - the cycle's electron-donor. This study demonstrates the malleability of metabolism and its capacity to switch trophic modes using only a small number of genetic changes and could facilitate transforming other heterotrophic organisms into autotrophs.

eLife assessment

This is an **important** follow-up study to a previous paper in which the authors reconstituted CO₂ metabolism (autotrophy) in *Escherichia coli*. Here, the authors define a set of just three mutations that promote autotrophy, highlighting the malleability of *E. coli* metabolism. The authors make a **convincing** case that mutations in *pgi* are loss-of-function mutations that prevent metabolic efflux from the reductive pentose phosphate autocatalytic cycle, and their data suggest possible roles of mutations in two other genes - *crp* and *rpoB*. This research will be particularly interesting to synthetic biologists, systems biologists, and metabolic engineers aiming to develop synthetic autotrophic microorganisms.

Introduction

Anthropogenic interference and intensive burning of fossil fuels releases large amounts of sequestered carbon in the form of CO₂, a main driver of climate change (Olofsson and Hickler 2008; Friedlingstein et al. 2022). Direct air carbon capture, by any means, is urgently needed to help reduce emissions. Here, we focus on biology-based carbon fixation as a potential avenue for CO₂ capture, which has been shown to be a potentially efficient production platform (Leger et al. 2021; Van Peteghem et al. 2022; Schwander et al. 2016). Specifically, we suggest using microbes for CO₂ valorization and produce a diverse repertoire of products including food, fuel, bio-plastics or other commodities (Claassens et al. 2019; Gleizer et al. 2020; Li et al. 2012; Nielsen and Keasling 2016). Natural autotrophs already metabolise CO₂ efficiently, yet most are challenging to cultivate and manipulate genetically compared to heterotrophic model organisms like *E. coli*. Here we explore an emerging alternative approach – converting heterotrophic model organisms into tractable autotrophs for bio-production (Gleizer et al. 2020; Baumschabl et al. 2022; Gassler et al. 2020; Gleizer et al. 2019). In addition to promising engineering applications, studying the transition from heterotrophic to autotrophic metabolism can uncover fundamental principles about the structure and regulation of carbon fixation metabolism (Gleizer et al. 2019; Gassler et al. 2020; Antonovsky et al. 2016; Barenholz et al. 2017; Flamholz et al. 2020; Herz et al. 2017).

Recently, with the increasing interest in platforms for sustainable production, synthetic biologists have been able to successfully manipulate different organisms to enable major metabolic transitions (Gassler et al. 2020; Gleizer et al. 2019; Kim et al. 2020; Chen et al. 2020; Keller et al. 2022; Bang et al. 2020; Satanowski et al. 2020; Yishai et al. 2018). These efforts resulted in the successful introduction of non-native C1 utilisation pathways, and even included successful transitions from heterotrophic into completely autotrophic metabolisms in bacteria (Gleizer et al. 2019) and yeast (Gassler et al. 2020). These previous studies introduced the reductive pentose phosphate (rPP) cycle, the most prominent carbon fixation pathway in nature (also known as the Calvin-Benson cycle), into *Escherichia coli* (*E. coli*) and *Komagataella phaffii* (previously *Pichia pastoris*). Indeed, we showed that *E. coli* was able to utilise the rPP cycle for the synthesis of all biomass carbon - thus converting it from a heterotroph to an autotroph (Gleizer et al. 2019). However, this transition required continuous culture in selective conditions for several months (adaptive lab evolution) and resulted in many uninterpretable mutations, exposing a large knowledge gap.

Using adaptive lab evolution presents a challenge as it is hard to isolate the changes necessary for a given phenotype from the overall set of accumulated mutations. Here we aimed to characterise the landscape of mutations required for autotrophic growth. Studying mutations that arise in multiple lineages can help us better understand the biological basis for the heterotroph-to-autotroph transition. To achieve this goal, we created a pipeline that enabled us to distinguish mutations essential for autotrophy from unspecific low-frequency mutations, i.e. ones that are generally promoted by the physiological conditions in evolution experiments. The final outcome of this pipeline is a compact essential set of changes permitting autotrophy in *E. coli*. We then proceeded with further analyses to shed light on the mechanism behind this compact set of essential mutations. We show the phenotype included a change in the activity of a central carbon metabolism enzyme and intracellular changes of the non-native carbon fixation cycle's metabolite and co-factors pools. We conclude by speculating about the role of these metabolic changes.

Results

A compact set of mutations that enable autotrophic growth was identified using rational design, iterative lab evolution and genetic engineering

Adaptive lab evolution is an effective tool that enables integration of non-native metabolic pathways. In order to harness the power of adaptive lab evolution, a selection system must be implemented to direct the evolution towards the desired function. In our case, the selection for the activity of the non-native rPP cycle. To ensure that some level of carbon fixation by RuBisCO becomes essential for growth, we knocked out phosphofructokinase (*pfkA* and *pfkB*), and 6-phosphate-1-dehydrogenase (*zwf*) which creates a stoichiometric imbalance and growth arrest when growing on five carbon sugars such as xylose (Gleizer et al. 2019 [\[1\]](#)). This imbalance could be rescued by a metabolic bypass - the phosphorylation of ribulose-5-P by Prk and the subsequent carboxylation by RuBisCO, thereby coupling growth to the heterologously expressed rPP enzymes. Critically, this setup allowed us to evolve *E. coli*, a natural heterotroph, into an autotroph, using a non-native rPP cycle to utilise CO₂ as a sole carbon-source. This system also requires formate dehydrogenase to oxidise formate for energy (Gleizer et al. 2019 [\[1\]](#)).

Although adaptive lab evolution is a powerful tool, it also presents a challenge - once a new phenotype has been achieved, it is hard to separate the essential changes required for its appearance from the overall set of accumulated mutations. In many cases, mutations occur on the background of highly adaptive ones, leading to a spread of neutral “hitchhiker” mutations in the population. In other cases, mutations that appeared at the beginning of the evolution are no longer needed for the final phenotype. Our goal was to find a set of mutations that, if introduced into a heterotrophic wild type *E. coli*, would result in the successful integration of the non-native carbon fixation and energy modules. This would enable autotrophic growth based solely on genetic engineering without the need for further evolutionary adaptation and potentially uncovering the mechanism behind this metabolic transition.

To isolate the essential genetic changes, we used a workflow that included three main stages as shown in **Fig. 1** [\[1\]](#). A) rational design-introduction of required heterologous genes and metabolic knockouts to enforce RuBisCO-dependent growth; B) adaptive lab evolution-revealing mutation candidates by growing the designed strain in autotrophic-selecting conditions over many generations until an autotrophic phenotype is obtained; C) evolution-inspired genetic engineering-introducing the most promising mutations revealed in stage B into the designed strain and testing it for autotrophic growth. Stages B and C were repeated until no further evolution was required, i.e. the designed strain in stage C had an autotrophic phenotype.

To find the most promising mutations in stage B, we applied at least one of two criteria: mutations in genes that occur repeatedly in different autotrophic lineages (but are uncommon in other adaptive lab evolution experiments) or mutations that when reverted, result in a loss of phenotype. We discovered these “consensus” mutations by sequencing isolated autotrophic clones, obtained from repeated adaptive lab evolution experiments-similar to the conditions we used in Gleizer et al. 2019 [\[1\]](#) (**Fig. 1** [\[1\]](#); steps B1, B2).

In practice, we inoculated the strain denoted as Ancestor into a chemostat with limited xylose and excess formate. After ≈3 months (≈60 chemostat generations) we isolated from the chemostat a clone able to grow under autotrophic conditions (methods), and compared its mutation set to those of two previously evolved autotrophic clones (Gleizer et al. 2019 [\[1\]](#)). The intersection of the three clones contained 4 genes that were mutated in all autotrophic isolated clones: the central carbon metabolism enzyme phosphoglucosomerase (*pgi*), the beta subunit of the RNA polymerase (*rpoB*), poly A polymerase (*pcnB*) and the MalT DNA-binding transcriptional activator (*malT*) (**Fig.**

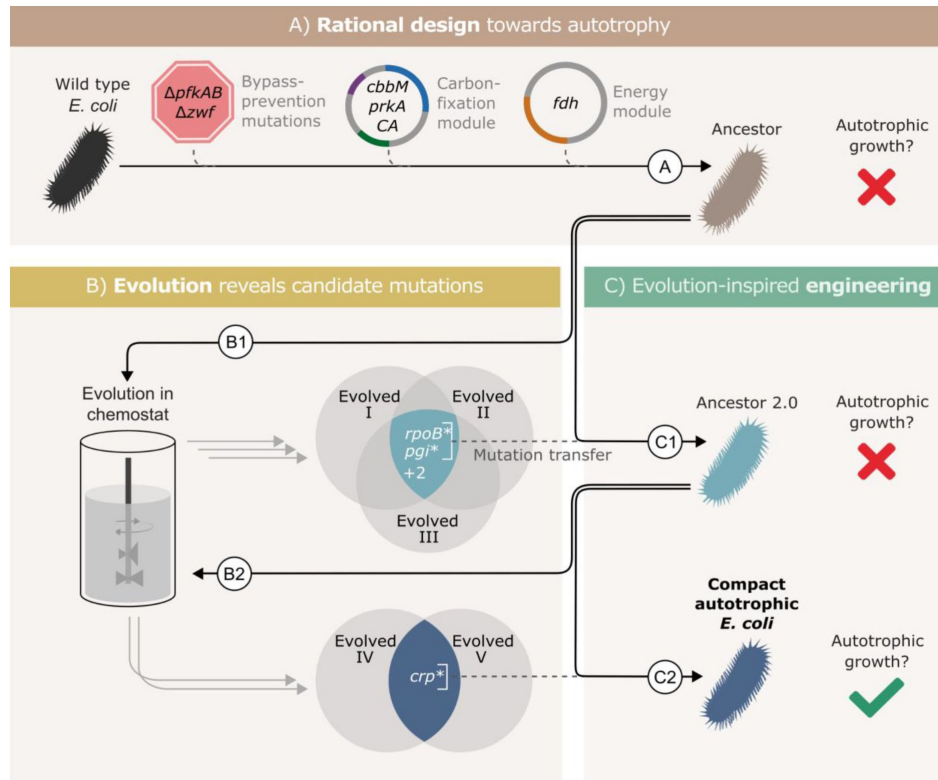


Fig. 1.

Autotrophic phenotype achieved by introducing 3 mutations on top of a rationally designed ancestor.

(A) We rationally designed the “wild-type” *E. coli* background strain (BW25113, depicted as a black bacterium) by introducing 4 enzymes: RuBisCO (*cbbM*), phosphoribulokinase (*prkA*), carbonic anhydrase (*CA*) and formate dehydrogenase (*fdh*), and 3 genomic knockouts: glucose-6-phosphate dehydrogenase (*zwf*) and phosphofructokinase A&B (*pfkAB*). We denote the resulting strain as “ancestor” (brown bacterium). **(B)** We tested the resulting strain for autotrophic growth and, since it didn’t grow, we used it for adaptive lab evolution in xylose-limited chemostats - i.e., conditions that select for higher carbon fixation flux. Altogether, we were able to isolate three evolved clones (two distinct strains from one chemostat experiment, Evolved I & II, and another strain from a second experiment, Evolved III, step B1). **(C)** Out of four consensus mutations that we identified, two - *rpoB* and *pgi* - were incorporated into the ancestor, giving “ancestor 2.0” (light blue bacterium, step C1). Once more, we tested for autotrophic growth and were unsuccessful. Therefore, we initiated another round of adaptive lab evolution experiments using ancestor 2.0 growing in two xylose limited chemostats and isolated two evolved clones (one from each chemostat, step B2). The two clones shared a single consensus mutation, in *crp*. We thus created a new strain from ancestor 2.0 by introducing the *crp* mutation into it (blue bacterium, step C2). This strain could grow autotrophically and thus we achieved a compact autotrophic strain.

1 [↗](#), step B1). *rpoB*, *pcnB* and *malT* are genes that are commonly mutated in adaptive lab evolution experiments (Phaneuf, Gosting, and Pálsson 2019 [↗](#)), suggesting that they have little relevance for autotrophic growth. Therefore, we chose to focus on the *pgi* mutation.

We introduced the H386Y *pgi* mutation into the rationally designed ancestor (**Fig. 1** [↗](#), step C1), and the engineered strain was tested for growth in autotrophic conditions (methods). No growth was observed in those conditions, which meant that additional mutations were necessary in order to achieve autotrophy.

Genomic sequencing revealed that during the genetic manipulation for introducing the *pgi* mutation, another mutation in *rpoB* (A1245V) appeared in the genome. In parallel, reverting a variety of RNA polymerase mutations in other evolved strains back to their wild-type allele, showed that mutations in RNA polymerase are in fact essential to the phenotype. Therefore, despite the fact that it was unintentional, and different from the other *rpoB* mutations observed in the ALE experiments, we left the mutation in the genome and proceeded with this strain.

This strain, mutated in both *pgi* and *rpoB* (denoted “ancestor 2.0”) was then used as the starting point for another round of evolution (**Fig. 1** [↗](#), step B2). Since it had already contained two mutations that were likely to be relevant for the desired phenotype, we expected to observe evolved autotrophic strains after fewer generations, compared to the first round, and to find fewer mutations in them. Indeed, on two separate experiments, it took roughly one month until autotrophic growth was observed, as opposed to more than 90 days in previous attempts. Even though these strains (**Fig. 1** [↗](#), “Evolved 4” and “Evolved 5”) both had multiple mutations, only one mutation was shared between these independent evolution experiments - a non-synonymous mutation in the cAMP receptor protein, *crp* (H22N). When we introduced this mutation to the “ancestor 2.0” background in the evolution-inspired genetic engineering stage (**Fig. 1** [↗](#), step C2), the strain was immediately able to grow under autotrophic conditions.

Since the RNA polymerase mutation was introduced into this strain (*rpoB* A1245V) as a byproduct of genetic engineering and never observed in the adaptive evolution experiments, we needed to verify its essentiality to the autotrophic phenotype. We found that a reversion of *rpoB* A1245V back to the wild-type allele causes a loss of the autotrophic phenotype. We therefore could conclude that three mutations (*pgi**, *crp** and *rpoB**) are sufficient to facilitate the autotrophic phenotype on top of the heterotrophic ancestor background. We denote the final engineered autotrophic *E. coli* strain, the “compact autotrophic *E. coli*”.

We verified the genotype using whole genome sequencing (methods). The results included the rationally designed knockouts ($\Delta pfkA$, $\Delta pfkB$, Δzwf), heterologous plasmids (energy and carbon-fixing modules) and the three introduced mutations (*pgi**, *crp** and *rpoB**). During the genetic engineering process, two additional mutations occurred unintentionally in the genes *uhpT* and *yejG*. *uhpT* is a hexose transporter and is unlikely to have any effect on the autotrophic phenotype, especially because the mutation was an early stop codon (Q7*) and likely a loss of function. As the function of *yejG* is unknown, we wanted to ensure that it is not essential to the autotrophic phenotype. Therefore, we reverted the mutation back to its wild-type allele and found that indeed the cells were still autotrophic.

Thus, we constructed a new autotrophic strain with three essential mutations (*pgi*, *crp* & *rpoB*), which we denoted the *compact* strain. When comparing this compact strain to a previously characterised autotrophic strain (Gleizer et al. 2019 [↗](#)), it was nearly identical in terms of growth rate, lag time and yield (**Fig. 2A** [↗](#)). Using ^{13}C labelling, we verified that all the biomass carbon stems from CO_2 (**Fig. 2B** [↗](#)). We termed the three introduced mutations (*pgi*, *crp* & *rpoB*) together with the carbon fixation machinery (*cbbM*, *prkA* & *CA*) and the energy module (*fdh*) the “autotrophic enabling gene set”.

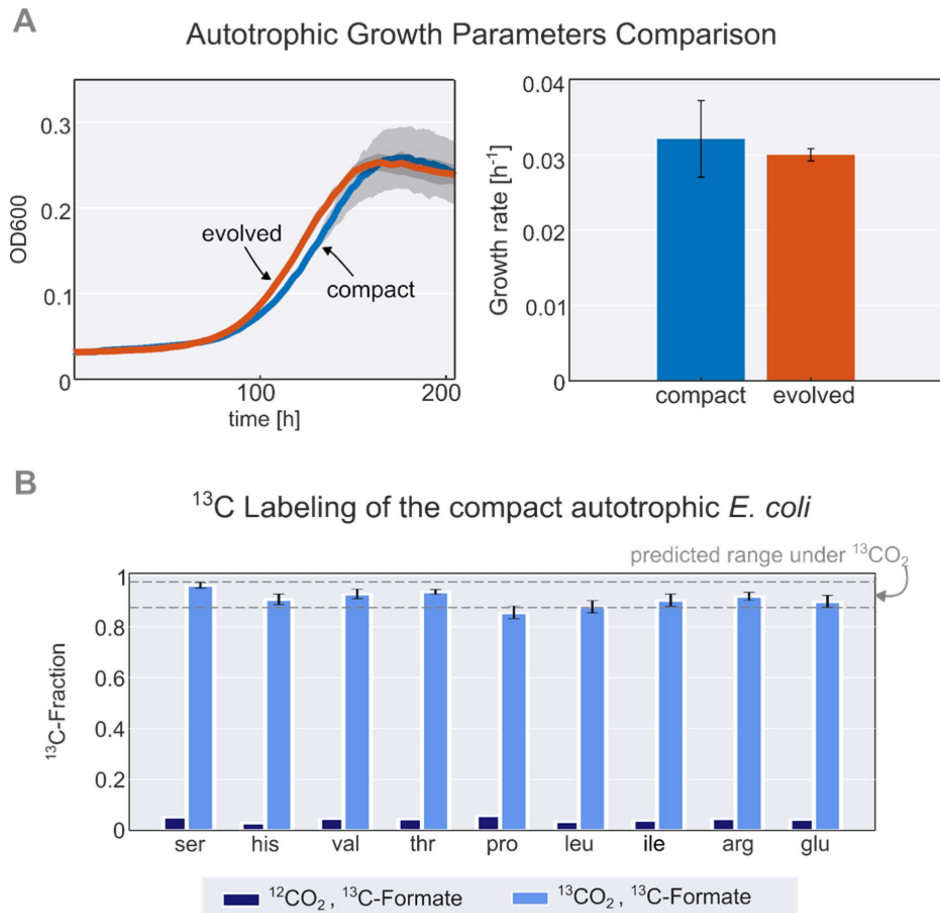


Fig. 2.

Validation and characterisation of the autotrophic phenotype.

(A) left: growth curve of an isolated evolved clone vs the engineered compact strain in liquid M9 minimal media supplemented with 45mM sodium formate and sparged with a gas mixture of 10% CO₂, 5% oxygen, and 85% nitrogen. right: calculated growth rate using linear regression of the engineered compact strain (light blue) and evolved strain (orange), the calculated doubling time with the given conditions is about 24h ($\mu=0.03\text{h}^{-1}$). Growth was carried out in triplicates (n=3) in a Spark plate reader with gas control, dark grey area corresponds to the standard deviation of the mean (150 μl culture + 50 μl mineral oil, to prevent evaporation). **(B)** The fractional contribution of ¹³CO₂ to various protein-bound amino acids of the compact autotrophic strain after ≈4 doublings on ¹³CO₂ and ¹³C labelled formate (light blue, mean of n=3). This reached the expected ¹³C labelling fraction (see methods) of the biosynthesized amino acids. When grown in naturally labelled CO₂ and ¹³C labelled formate (dark blue, n=1, technical triplicate) the ¹³C content dropped to close to natural abundance. Experiments with ¹³CO₂ as the substrate were carried out in air-tight (i.e., sealed) growth vessels.

Introduction of autotrophy-enabling gene set into wild-type background does not require bypass-prevention for growth

As described above, in order to select for autotrophic growth, the native *E. coli* was metabolically re-wired by 3 auxiliary genomic knockouts ($\Delta pfkA$, $\Delta pfkB$, Δzwf). These knockouts created a dependency on carboxylation by RuBisCO for growth even when consuming pentose sugars which were supplemented during the chemostat evolution (see methods, Antonovsky et al. 2016 [↗](#); Gleizer et al. 2019 [↗](#)). This dependency was expected to direct the evolution towards increased usage of the synthetic rPP cycle and, eventually, to autotrophic growth.

Once autotrophic growth was achieved, and the enabling gene set was found, we wanted to test if these auxiliary knockouts (*pfkA* *pfkB* and *zwf*) were still required for the phenotype. We introduced the set of autotrophy-enabling genes into a wild-type (BW25113) *E. coli* background (methods, Fig. 3 [↗](#)). After doing so, we discovered an unintentional genomic mutation, in the ribosomal 16S subunit (*rrsA*) that was introduced during the genomic manipulation process. The engineered *E. coli* strain was able to grow autotrophically, although its final OD was half and its growth rate was less than half of that of the rationally designed compact strain still harbouring the auxiliary knockouts (Fig. S2 [↗](#)). This indicates that the “autotrophic enabling gene set” is sufficient to achieve the trophic mode change and that the knockouts are helpful but not essential for the final autotrophic growth. As discussed below, this demonstrates that, in some cases, metabolic rewiring could be viewed as a temporary “*metabolic scaffold*” which can be removed from the genotype once the desired phenotype is achieved.

We thus established a new engineered autotrophic *E. coli* strain containing a set of three essential and sufficient mutations, and showed that the bypass-prevention knockouts introduced before the ALE are themselves not necessary for the phenotype. Due to the small required number of genetic modifications, this strain provides an opportunity to derive guidelines for future engineering efforts that use the rPP cycle. Having this goal in mind, we focused our attention on identifying the relevant phenotypes of each of these mutations and explaining their adaptive advantage. Because the mutations were essential for the autotrophic growth, we could not use autotrophic conditions to compare them to a strain with wild-type alleles. Therefore, we chose the most suitable combinations of strains and conditions that could isolate the effect of the pertinent mutation.

Modulation of a metabolic branch-point activity increased the concentration of rPP metabolites

Metabolic pathways that regenerate and synthesise more of their own metabolites are referred to as autocatalytic cycles. Within the autotrophic *E. coli* the rPP cycle is autocatalytic. Due to the inherent positive feedback mechanism, autocatalytic cycles tend to be unstable, and therefore the fluxes of entry and exit points (bifurcation/branch points) need to be balanced (Barenholz et al. 2017 [↗](#)). Such tuning is needed since any disruption of the balance between cycling and branching fluxes could result in the depletion, or alternatively toxic accumulation of intermediate metabolites, and cycle arrest.

We previously predicted that mutations in branch points will be needed to stabilise the steady state flux within the rPP cycle (Barenholz et al. 2017 [↗](#)). In line with this prediction we find that the mutation in phosphoglucosomerase (Pgi), a key branch point in the rPP cycle, follows this design principle. Pgi consumes fructose-6-phosphate (F6P), one of the cycle intermediates, converting it into glucose-6-phosphate (G6P), a precursor for cell membrane biosynthesis. Thus, Pgi regulates flux out of the autocatalytic cycle (Fig. 4A [↗](#)).

In the different adaptive lab evolution experiments, we found three distinct mutations in the *pgi* gene. The first, H386Y, is a non-synonymous mutation occurring in one of the catalytic residues in the active site, and is part of the autotrophic enabling gene set. The second was a complete

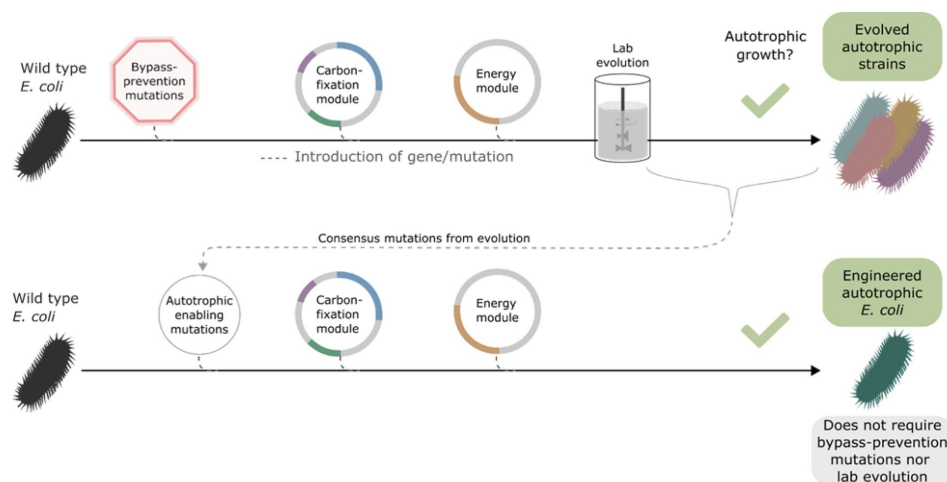


Fig. 3.

Auxiliary knockouts are not required for final phenotype.

We transformed a wild-type BW25113 *E. coli* strain (black bacteria) with the carboxylating and energy module plasmids (grey circles with coloured gene annotations), and inserted 3 auxiliary genomic knockouts (red octagon) to rewire metabolism toward carboxylating dependency. This strain was used for iterative evolution experiments in order to generate diverse autotrophic strains and reveal mutations candidates for rational design, i.e. consensus mutations. The identified autotrophic enabling mutations (thin grey circle) were then introduced in a wild type strain expressing the heterologous plasmids but without the auxiliary knockouts, the final strain was able to grow in autotrophic conditions. Dashed lines represent gene/mutation introduction.

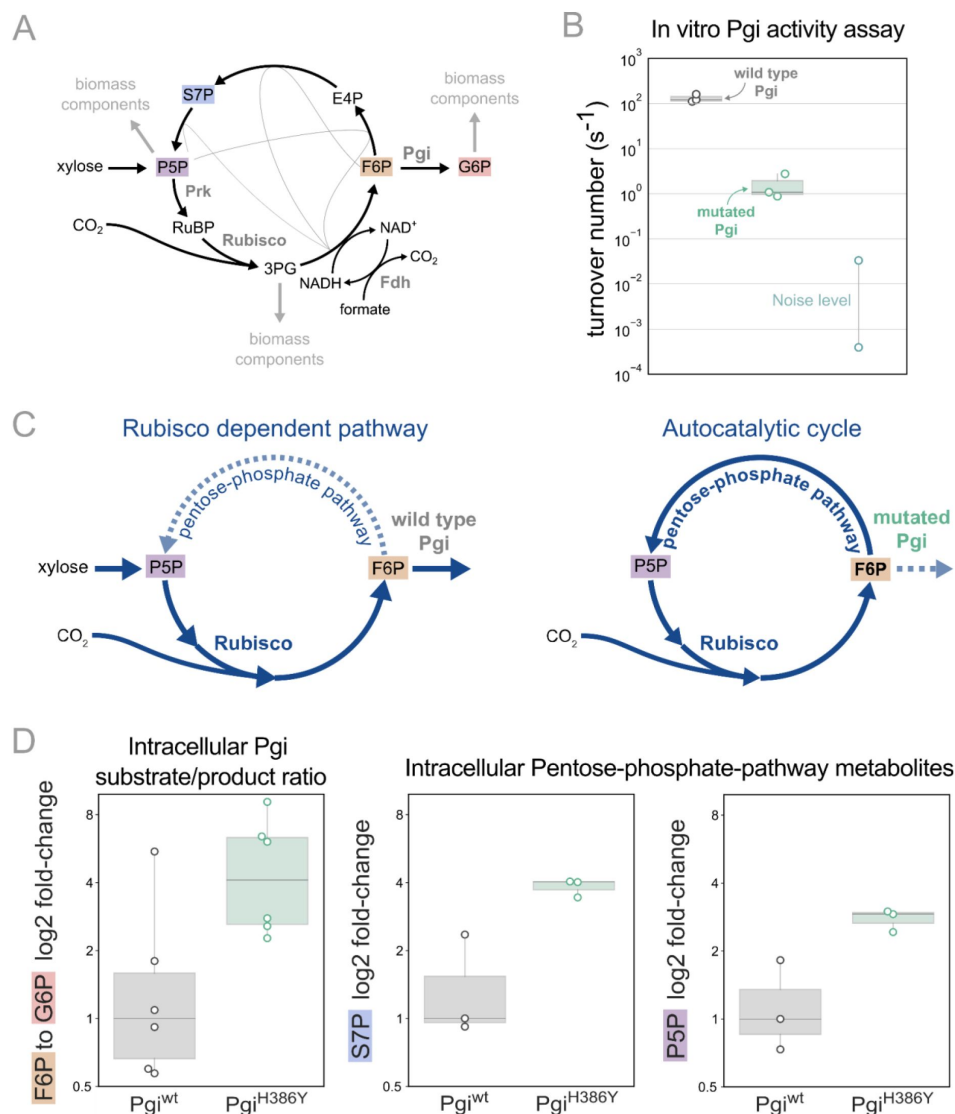


Fig. 4.

Pgi mutation is required to partition flux towards regeneration of autocatalytic cycle substrate.

(A) Metabolic scheme of rPP cycle components when growing on xylose. **(B)** In vitro spectrophotometric coupled assay determined that the rate of the isomerization of fructose-6-phosphate (F6P) to glucose-6-phosphate (G6P) is ≈ 100 -fold lower for purified Pgi^{H386Y} (≈ 1.5 [s⁻¹]) compared to Pgi^{wild-type} (≈ 130 [s⁻¹]), $n=3$. Noise level was determined by negative control samples containing a non-Pgi enzyme. **(C)** Left: The wild type pgi, competes for its substrate F6P with the autocatalytic cycle, resulting in low F6P pool and low regeneration rate of ribulose-5-phosphate (Ru5P), which means that the RuBisCO-dependent pathway requires constant xylose supply. Right: The mutated Pgi (green) has reduced activity, which increases the ratio between the flux in the cycle and efflux which is needed for a stable regenerative flux towards Ru5P (via the pentose-phosphate-pathway), thus enabling an autotrophic cycle. **(D)** Left: Measured relative intracellular ratio of F6P and G6P of the ancestor background harbouring a Pgi^{H386Y} mutation compared to the ancestor with a wild-type Pgi, both grown in RuBisCO dependent conditions, $n=6$ cultures p -value < 0.05 . Right: Relative intracellular abundance of pentose-phosphate-pathway metabolites - sedoheptulose-7P (S7P) and total pool of pentose phosphates (ribulose-5P, ribose-5P, ribose-1P, and xylulose-5P - denoted P5P) in a Pgi^{H386Y} strain versus the Pgi^{wild-type} strain, both growing in a RuBisCO-dependent manner, $n=3$ cultures.

knockout, a 22KB chromosomal deletion, including also 16 other genes. The third was an early stop codon E72*. These observations led us to suspect the H386Y mutation in *pgi* decreases or even completely eliminates the activity of the enzyme.

We tested the kinetic rates of the isomerization of F6P to G6P of two purified Pgi enzymes, wild type Pgi and Pgi H386Y, using a spectroscopic coupled assay, where G6P was coupled to NADP⁺ reduction (Widjaja et al. 1998 [↗](#)) (**Fig. 4B** [↗](#); see also methods). The measured k_{cat} of the wild type Pgi was $\approx 130 \text{ [s}^{-1}\text{]}$ (± 26) which is in line with previous computational predictions of a k_{cat} of $\approx 200 \text{ [s}^{-1}\text{]}$ (Davidi et al. 2016 [↗](#)). The mutated Pgi showed weak activity with a k_{cat} of $\approx 1.5 \text{ [s}^{-1}\text{]}$ (± 1). Therefore the *pgi* mutation is in line with the original expectation that branching point regulation is necessary, and the nature of the observed mutations in *pgi* supports the notion that their role is to reduce the efflux from the non-native autocatalytic cycle.

Pgi is part of glycolysis/gluconeogenesis, where significant flux is required in wild type *E. coli*. Accordingly, wild type Pgi catalyses a reaction that is causing a strong efflux of F6P from the cycle, diminishing the regeneration of ribulose-1,5-bisphosphate (RuBP) which is usually not needed in the native metabolism but is essential for the rPP autocatalytic activity. In the mutated version of Pgi the efflux capacity is diminished, which can stabilise the cycle (**Fig. 4C** [↗](#)).

Following the same logic, we can expect an increase in the F6P substrate metabolite pool due to reduced consumption by Pgi. Therefore, we measured intracellular sugar-phosphates in the *pgi* mutant strain and a strain with a wild-type *pgi* allele by liquid chromatography-tandem mass spectrometry (LC-MS/MS). For this comparison, we used the RuBisCO-dependent ancestor strain as the genetic background. We found that the ratio of F6P to G6P was about 3 times higher in the *pgi* mutant strain relative to the wild-type (**Fig. 4D** [↗](#)). Furthermore, the *pgi* mutant had higher levels of metabolites within the rPP cycle (**Fig. 4D** [↗](#)), confirming the stabilising function of the mutation.

These results lead us to conclude that the recurring mutations in the *pgi* gene are mediating the integration of the non-native carbon fixation genes into a stable autocatalytic cycle. This is achieved by reducing the ratio of the efflux from F6P to G6P relative to the flux in the cycle, thereby redirecting it towards the rPP cycle and the regeneration of RuBP.

Proteomic analysis reveals up-regulation of rPP cycle and formate-associated genes alongside down-regulation of catabolic genes

The other two mutations in the compact set - *crp* (H22N) and *rpoB* (A1245V), are global effectors in the cell. Crp is one of the most general transcription factors in the cell and has a role in many processes from metabolism to osmoregulation and biofilm formation (Landis, Xu, and Johnson 1999 [↗](#); Jackson, Simecka, and Romeo 2002 [↗](#); Ammar, Wang, and Rao 2018 [↗](#)). The position H22 in Crp is part of “activation region 2” of the protein, an interaction region with RNA polymerase (**Fig. S8** [↗](#)). This position was shown to have an effect on promoter specificity (Niu et al 1996 [↗](#)) and was previously reported to have a role in the integration of the rPP cycle in *E. coli* (Herz et al. 2017 [↗](#)).

Similarly, mutations in *rpoB*, a subunit of RNA polymerase, have been shown to have a large effect on the cell transcriptome (Conrad et al. 2010 [↗](#); Utrilla et al. 2016 [↗](#)). The observed A1245V mutation in *rpoB* is situated within the core of RNA polymerase, proximal to its interaction with the DNA (figure S8). Furthermore, this genetic variation was previously documented in the adaptation of the *E. coli* B REL1206 strain to high temperatures (Tenaillon et al. 2012 [↗](#)) and is closely positioned to another previously reported *rpoB* mutation (K1242Q) that appeared in adaptation to loss of Pgi activity (Charusanti et al. 2010 [↗](#)).

Depending on the transcribed gene target, the effect of the two mutations might be additive, antagonistic, or synergistic. Since each one of these mutations individually (in combination with the *pgi* mutation) is not sufficient to achieve autotrophic growth, it is reasonable to assume that only the target genes whose levels of expression change significantly in the double-mutant are the ones relevant for the autotrophic phenotype. Given the accumulated knowledge described above, these mutations are likely to have a broad effect on the cell. We chose to further elucidate their effect using a proteomics approach. As such, we conducted two experiments in an attempt to find changes in expression that are relevant for the adaptation to the new trophic mode. The first experiment compared the proteome of an evolved clone to its ancestor. The second experiment similarly compared an engineered BW25113 (BW) strain, containing the two regulatory mutations from the compact set (i.e., *crp* H22N and *rpoB* A1245V), to a wild type BW strain.

Since our aim was to observe only direct effects induced by the genetic changes, rather than by the environment, we measured the proteomes of the strains under identical conditions and dilution rates. Ideally, the comparison should be done in conditions that require autotrophic growth (since some effects might depend on the lack of an organic carbon source), however, all strains except the evolved clone cannot grow in such conditions. Therefore, we decided on growing the cells in the conditions present at the beginning of the lab evolution experiment, namely a xylose-limited chemostat. The dilution rate was set to $\approx 0.03 \text{ h}^{-1}$ leading to a doubling time of $\approx 24 \text{ h}$ (methods).

In order to identify the genes with a large and significant fold change (fold change >4 , p-value <0.05) we generated a volcano plot for each such comparison (**Fig. S6**). We then proceeded by comparing the output of the first experiment (evolved divided by ancestor, **Fig. S6A**) to that of the double mutant divided by the base BW strain (**Fig. S6B**), since both mutations are essential for the phenotype and are known to affect expression. We focused on genes with common trends of expression level in both sets (**Fig. 5**).

The proteomics analysis yielded few genes that had an upregulation of >4 -fold in both the evolved versus ancestor and the double mutant versus BW base strain experiments. Two of these up-regulated genes, *fdoG* and *fdoH*, are subunits of the native formate dehydrogenase O enzyme complex (FDH-O) and are therefore related to the energy module. These genes were upregulated in both the evolved strain and in the strain with the *crp* H22N and *rpoB* A1245V double mutation (≈ 5 -fold increase in both cases). The FDH-O enzyme complex takes part in the respiratory chain formed by formate dehydrogenase and dimethyl sulfoxide (DMSO) reductase. Here, the transfer of electrons from formate to DMSO (via a quinone electron carrier) is coupled to the generation of a proton-motive force across the cytoplasmic membrane for energy formation.

Another observed upregulated protein was ribose-phosphate-isomerase B (*rpiB*), which was up-regulated in the evolved strain (>40 -fold increase) and in the double mutant ≈ 5 -fold increase (**Fig. 5**). *rpiB* is an inducible isoenzyme of the constitutively expressed ribose-phosphate-isomerase A, a core enzyme in the rPP cycle. This might serve as evidence for a higher flux requirement through the pentose-phosphate-pathway, supported by expressing a second isoenzyme. Other explanations may include some form of regulation, perhaps allosteric, inhibiting RpiA activity, making the strain dependent on RpiB activity, or non-specific enzymatic activity of RpiB which is required by the cell. The fourth up-regulated gene - *ftsI*, a peptidoglycan DD-transpeptidase - does not have a clear explanation at this time.

At the other end of the expression spectrum, several pathways were down-regulated in both the evolved and in the BW double mutant strain (**Fig. 5**). Down-regulation of the starvation response, metabolic and PTS component genes seems to be at the heart of the adaptation to autotrophic growth, with several operons experiencing >4 -fold decrease in expression. Most notably the *gat*, *ara* and *man* operons responsible for galactitol, arabinose and mannose degradation, respectively. Again, this is evident in both the evolved and the double mutant strains (**Fig. 5**). Other highly down-regulated metabolic pathways include amino acid biosynthesis,

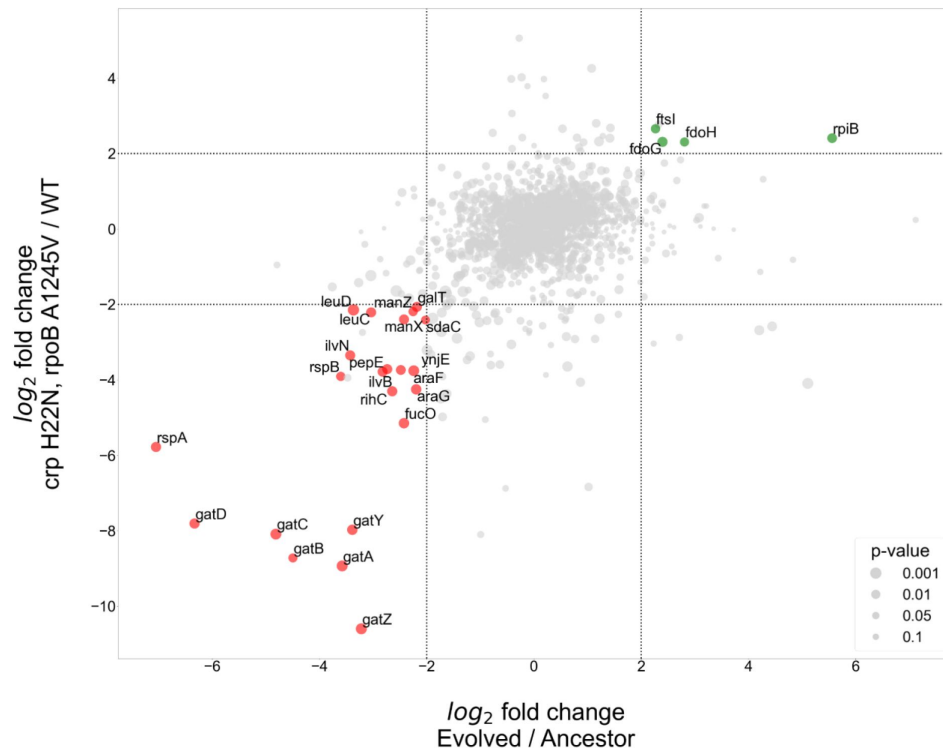


Fig. 5.

Proteomics analyses of fold change in evolved/ancestor expression levels versus double mutant/WT.

Comparison of the $\log_2$ fold change in the evolved autotrophic strain divided by the ancestor expression versus BW rpoB A1245V & crp H22N, divided by the WT. The size of dots represents significance, only genes with >4 fold-change who showed significance (<0.05 p-value) were annotated.

specifically valine and leucine, moreover we observed a down regulation in *rspA* & *rspB*, which encode for starvation sensing proteins. We suggest these results mean that many pathways usually up-regulated in *E. coli* during starvation may be redundant or even counter-productive in an autotrophic context, and decreasing their expression could result in increased fitness.

We additionally tested whether separating between the two regulatory mutations (*crp* and *rpoB*) might provide a clearer picture, e.g. if different up- or down-regulated genes are targeted by only one of these mutations and not the other. Therefore, we used single-mutant BW strains and compared them to the same BW wild-type as before. Essentially, the results did not change our conclusions and although in most cases the effect of the two mutations was cumulative, we did find at least one case where they were antagonistic (Fig. S7). For example, *fdoG* and *fdoH* showed a ~4-fold decrease compared to wild type when the *crp* mutation was introduced, but (as stated earlier) had a ~5-fold abundance increase in the double mutant, suggesting an epistatic effect between the *rpoB* and *crp* mutations (Fig. S6B & Fig. S7B). All the comparisons based on the single and double mutant strains can be found in the supplementary.

Mutations in regulatory genes lead to increased availability of the carbon fixation cycle electron donor NADH

Along with the effect on expression, we suspected the two transcription-associated mutations (*crp* & *rpoB*) to also have a metabolic and physiological effect. Since Crp and RpoB are both essential to the autotrophic phenotype and are known to indirectly interact in the cell (Savery, Rhodius, and Busby 1996; Niu et al. 1996; Frendorf et al. 2019), we address them as one unit. The proteomics analysis, where the strain with both mutations seem to have a mutual effect that is in some cases different than in the individual mutants, supported this decision. Mutations in *crp* and *rpoB* are known to affect growth at different conditions such as types of media or carbon sources (Utrilla et al. 2016; Conrad et al. 2010; Cheng et al. 2014; Frendorf et al. 2019). Therefore we tested whether this was also the case for the conditions under which the cells evolved in the chemostat.

We performed a growth experiment of the double mutant (*crp* H22N, *rpoB* A1245V) expressing *fdh* in a minimal medium containing xylose and formate. Under such conditions, the strain harbouring both mutations and *fdh* exhibited >25% increase in the final OD compared to the wild type expressing *fdh* (Fig. 6A & B). At the same time they also showed a longer (>10 hours) lag time. Notably, when grown solely on xylose, in the absence of formate, the increase in yield was not observed, and the extended lag was less pronounced (Fig. 6A).

Since NADH is the electron donor for carbon fixation, and a product of formate consumption via Fdh, we used metabolomics to measure the ratio of NADH/NAD⁺ in these strains in similar conditions (methods). All of the measured ratios were normalised to the wild-type (with *fdh*) growing in the absence of formate. The addition of 30mM formate, which can serve as an electron donor via oxidation to CO₂ by Fdh, increased NADH/NAD⁺ ratio by >20-fold in the double mutant, significantly higher than the ~10-fold increase in the wild-type strain (p-value <0.05). No significant difference was observed between the strains in the absence of formate (Fig. 6C).

NADH could affect growth in several ways, as discussed below. The specific reason for the increase in final OD or lag time remains inconclusive at this point but we hypothesise that it is related to the increased expression of Fdo observed in the proteomic analysis. We also note that as the proteomics results suggest - the mutations in global transcriptional machinery like Crp & RpoB could have many other effects on the autotrophic phenotype (Haverkorn van Rijsewijk et al. 2011) which await future exploration.

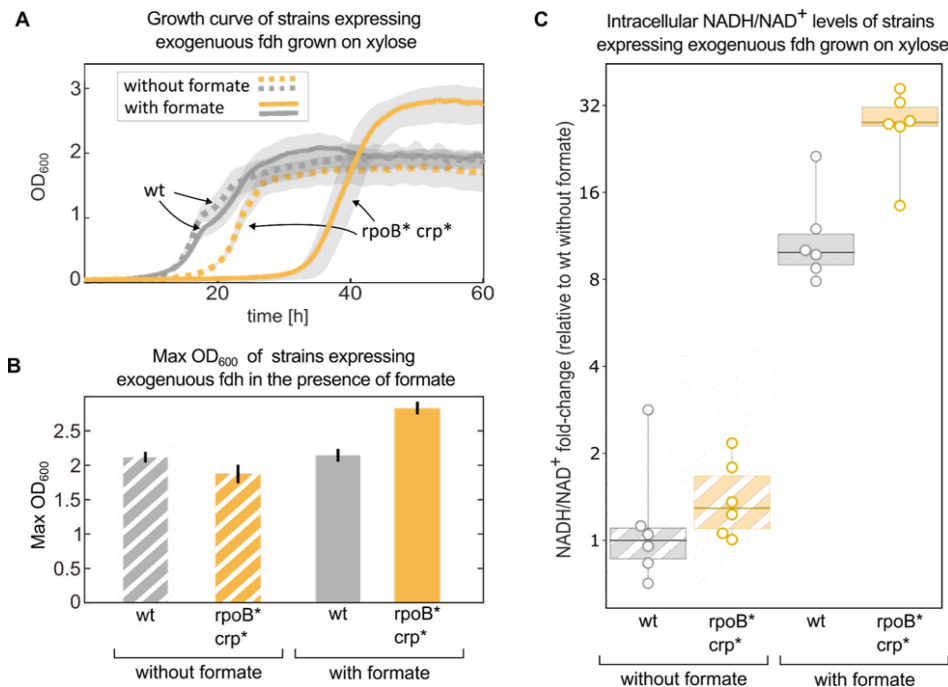


Fig. 6.

mutations in *rpoB* and *crp* increase yield and intracellular NADH/NAD⁺ levels in *fdh*-expressing *E. coli* in the presence of formate.

Growth experiment of BW25113 wild-type (grey) compared to a *crp* H22N *rpoB* A1245V mutant (orange). Both strains express *fdh*. (A) strains were grown in 1g/L xylose and 40mM formate (solid line) or in the absence of formate (dashed line). The experiment was executed in a 96 well plate in 10% CO₂ atmosphere, n=6 repeats. Lines represent the mean, light-grey background represents the standard deviation. (B) Maximal OD₆₀₀ of the wild-type (grey) and mutant (orange) strains grown in 1g/L xylose and 40mM formate or in the absence of formate. Bar heights represent averages (n=6) of the median of the top 10 OD measurements of each replicate. Error bars represent standard error. (C) Intracellular NADH/NAD⁺ ratio of the wild-type (grey) was compared to the mutant (orange). The strains were grown in 2g/L xylose and 30mM formate, or in the absence of formate. The y-values are NADH/NAD⁺ ratios as fold-changes relative to wild-type without formate. Boxes represent 25-75 percentile ranges and dark lines represent median values. All data points are depicted without removing outliers.

Discussion

In this study, we demonstrated how iterative evolution and engineering were able to narrow down the genomic mutations required for autotrophy in *E. coli* from tens of mutations to only three mutated genes. The three mutated genes - *pgi*, *crp* and *rpoB* - serve as a compact set, which when introduced into a wild-type background and supplemented with the carbon-fixing and energy modules, achieve an autotrophic phenotype. By metabolic, proteomic and physiological characterisation we were able to distil some of the effects of these mutations and understand their role in facilitating the autotrophic phenotype. We suggest that the three main effects achieved by the autotrophy-enabling mutations are: (i) integrating the non-native carbon fixation cycle by enhancing flux within the cycle over outgoing flux; (ii) facilitating the integration of the energy module via decoupling of formate harvesting into energy and redox, thereby increasing the NADH/NAD⁺ ratio, which could have both thermodynamic and kinetic benefits (iii) and freeing protein resources by decreasing unneeded expression.

The mutation in *pgi* affects the activity of its gene product phosphoglucosomerase - a metabolic enzyme that catalyses a reaction that branches out of the non-native rPP cycle. The enzyme converts F6P (a cycle intermediate) to G6P (a precursor required for membrane biosynthesis). In a previous study, this mutation was found to be essential for the integration of the rPP autocatalytic cycle and was part of a minimal set for hemi-autotrophic growth (Gleizer et al. 2019 [↗](#); Gassler et al. 2020 [↗](#); Antonovsky et al. 2016 [↗](#); Barenholz et al. 2017 [↗](#); Flamholz et al. 2020 [↗](#); Herz et al. 2017 [↗](#)). As our results suggest, the mutated enzyme shows a significant reduction in activity, which decreases the efflux from the rPP cycle to biosynthesis. We hypothesised that its role is to generate a new, higher, steady-state concentration for the rPP cycle metabolites, thus stabilising the cycle. This was supported by our observation that introducing a *pgi* mutation into a metabolically rewired RuBiCO-dependent strain resulted in increased concentration of rPP cycle intermediates. This exemplifies the key role branch-points play in autocatalytic cycles (Gleizer et al. 2019 [↗](#); Gassler et al. 2020 [↗](#); Antonovsky et al. 2016 [↗](#); Barenholz et al. 2017 [↗](#); Flamholz et al. 2020 [↗](#); Herz et al. 2017 [↗](#)), making them a leading target for rational attempts to integrate non-native autocatalytic cycles into metabolism.

The remaining two mutations act as global regulators, which presents a challenge in distinguishing their specific adaptive impacts from their broader effects on cellular expression patterns. Using proteomic analyses, we have managed to identify certain genes that had a major change in expression when one of the regulating mutations was introduced, and when both were introduced together.

One main observation was the down-regulation of several metabolic pathways including catabolism of alternative carbon sources and amino acid biosynthesis. We suggest this is an adaptive role in improving cell resource allocation, and a means to modulate resource allocation, possibly freeing up proteome resources from alternative carbon metabolism proteins. This includes proteins such as PTS system components or ATP dependent transporters, that tend to incur high energetic costs.

Other reasons these catabolic and transport pathways may result in reduced fitness could include competition over metabolites with the rPP cycle, or by causing other metabolic blunders such as dead end products with potential toxicity effects under the new flux regime. Another observed effect the *crp* mutation shows is the up-regulation of *rpiB*, a typically unexpressed isoenzyme to one of the reactions in the rPP cycle. This expression increase could be beneficial or even essential in a scenario where *rpiA*, the main enzyme catalysing this reaction, cannot support the flux required by the rPP cycle, for example due to kinetic capacity limitations or allosteric inhibition. Further work is needed to investigate if up-regulation of *rpiB* is indeed essential for the autotrophic phenotype.

We also suggest these mutations play a role in improving the utilisation of formate, which is supported by the up-regulation of *fdoG* & *fdoH*. We found that a strain introduced with both the *crp* and *rpoB* mutations demonstrated a pronounced up-regulation of over 4-fold and also a concrete fitness advantage in growth conditions which include formate. Metabolomics analysis showed this strain also has a significant increase in the NADH/NAD⁺ ratio. NADH is the essential cofactor for the electron transfer reaction of the non-native rPP cycle (catalysed by the D-glyceraldehyde-3P dehydrogenase GAPDH) and a product of the non-native formate-based energy module. The importance of this up-regulation is also supported by natural chemolithoautotrophic metabolism, which uses two distinct ways to harvest energy and reducing power from the electron-donor. For example, in the model chemolithoautotroph *Cupriavidus necator*, the oxidation of formate is catalysed by two distinct types of FDH: a soluble, NAD⁺-linked enzyme (S-FDH; EC 1.2.1.2) and a membrane-bound enzyme coupled directly to the respiratory chain (Friedrich, Bowien, and Friedrich 1979 [↗](#); Oh and Bowien 1998 [↗](#)).

We suspect utilising formate dehydrogenase O (FDH-O) in the periplasm is enabling the cell to also utilise formate directly on the membrane to create a proton motive force for energy production without the need to exhaust the intracellular NADH pool, thus allowing for its accumulation. Increased levels of NADH creates favourable thermodynamics for the rPP cycle flux and could also offer improved kinetics. This result is in line with previous models, predicting that high cofactor levels are required for a stable metabolic state of the rPP cycle (Janasch et al. 2019 [↗](#)). We suggest this observation offers an avenue for future rational design attempts.

We continue by hypothesising that such regulatory mutations are to be expected in adaptive laboratory evolution experiments, especially during selection for a major shift (novelty) in the phenotype. A major shift will often necessitate a modulation of the activity of many cellular components. Lab evolution is limited by population size and number of generations, which means it can only explore a small part of the genotypic space. Therefore, sets of three or more mutations, where each mutation on its own does not confer an advantage, have a negligible probability to occur. On the other hand, a single mutation in a global regulator could concurrently perturb the activity of hundreds of components and might be the only viable solution to capture several required changes simultaneously.

Another reported observation is how metabolic rewiring can serve as a “metabolic scaffold” to direct lab evolution. Since scaffold mutations are put in place for the purposes of selection, they can be removed once the goal is achieved. A common limitation of lab evolution is its tendency to follow evolutionary trajectories which increase fitness by deactivating or losing the heterologously expressed genes, thereby arriving at “dead-end” local optimum. A “metabolic scaffold” as utilised here can block these dead-end trajectories by coupling a target activity (here RuBisCO carboxylation) to growth, thereby increasing the likelihood of trajectories that lead to the desired phenotype. Once the phenotype is achieved, removal of the metabolic scaffold should be possible since growth is by definition dependent on the non-native genes. Using metabolic growth-coupling as a temporary “metabolic scaffold” that can be removed, could serve as a promising strategy for achieving minimally perturbed genotypes in future metabolic engineering attempts.

The approach presented here, of harnessing iterative lab evolution to find a compact set of essential mutations, strikingly shows the extreme malleability of metabolism and evolution capacity to switch trophic modes in laboratory time-scales with very few mutations. It can also serve as a promising pipeline for other genetic engineering attempts. Having a compact and well-defined genotype in a model organism allowed us to reveal metabolic adaptation steps needed for autotrophic growth, and could be used for future exploration of the design principles of the rPP cycle operation and its components (Aigner et al. 2017 [↗](#)). The harnessing and rationalisation of the lab evolution process and metabolic adaptation steps elucidated here could facilitate the introduction of the rPP cycle and other carbon fixation cycles into other organisms of interest.

Next steps could include integrating metabolic bio-production modules and alternative energy sources. Now that synthetic autotrophy in *E. coli* can be achieved with only a few defined steps, it expands the playground for more labs to join the quest for sustainable bio-production from CO₂.

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Data curation: RBN, EM, VP, BdP, EN

Methodology: RBN, EM, VP, BdP, GJ, SG, HL, EN and RM

Investigation: RBN, EM, VP, BdP, GJ, DL, HY, NN, DE and SG

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Writing – original draft: RBN, EM, EN and RM

Competing interests

We declare the following provisional patent related to the manuscript, “An Engineered Autotrophic *E. coli* Strain for CO₂ Conversion to Organic Materials.”

Data and materials availability

All data are available in the main text or the supplementary materials.

Supplementary Materials

Materials and Methods

Supplementary Text

Figs. S1 to S8

Data S1 to S17

Materials and Methods

Strains

We generated an engineered ancestor strain for chemostat evolution based on the *Escherichia coli* BW25113 strain (Grenier et al. 2014 [↗](#)). We used P1 transduction (Thomason, Costantino, and Court 2007 [↗](#)) to transfer knockout alleles from the KEIO strain collection (Thomason, Costantino, and Court 2007 [↗](#); Baba et al. 2006 [↗](#)) to our engineered strain, and to knock out the genes phosphofructokinase (*pfkA* and *pfkB*), and 6-phosphate-1-dehydrogenase (*zwf*). Following the transduction of each knockout allele the KmR selection marker was removed by using the FLP recombinase encoded by the pCP20 temperature-sensitive plasmid (Cherepanov and Wackernagel 1995 [↗](#)). Loss of the selection marker and the temperature sensitive plasmid were validated by replica-plating the screened colonies and PCR analysis of the relevant loci. The cells were then transformed with the pCBB plasmid(13) (accession number KX077536) and a pFDH plasmid with a constitutive promoter controlling the expression of the *fdh* gene that resulted from Gleizer et al. 2019(12). The transfer of mutations from evolved clones for the generation of ancestor 2.0, and other mutated strains was also accomplished by P1 transduction as reported in Herz et al. 2017 [↗](#). P1 lysate was prepared from Keio collection strains that had a Kanamycin (Km) resistance marker located near the desired allele (as specified in Supplementary Data S1). The strains carrying the desired mutated allele were transduced with P1 phages containing the resistance marker and then plated on Km plates. Donor strains were selected based on the presence of the desired mutated allele and the Km resistance marker in their colonies. A P1 lysate from each donor strain was used to transfer the desired mutations to a recipient strain. Because the transferred DNA fragment to

the recipient was relatively large (approximately 100 kb), including the selection marker, it often carried adjacent genes. Genotyping was performed using Sanger sequencing. For further genomic modification, pCP20, a temperature-sensitive plasmid encoding the FLP recombinase from the Clone and Strain Repository unit at WIS, was used following standard procedures. This allowed for the removal of Km resistance markers, enabling iterative rounds of genome editing.

Plasmids

To create the pFDH plasmid, an *E. coli* codon optimised DNA sequence of formate dehydrogenase from the methylothrophic bacterium *Pseudomonas sp. 101* (Popov and Lamzin 1994 [\[1\]](#)) was cloned with an N-terminal His-tag into a pZ plasmid (Expressys, Germany) under a constitutive promoter and with a strong ribosome binding site (rbs B of (Zelcbuch et al. 2013 [\[2\]](#))). The plasmid has a pMB1 origin of replication and therefore is present in high copy number. We replaced the KmR selection marker on the plasmid with a StrepR marker. An 8 bp deletion appeared in the promoter region of the first evolved clone. This plasmid was isolated and was the plasmid used for all consequential evolution experiments and autotrophic strains (Gleizer et al. 2019 [\[3\]](#)). Details regarding the pCBB plasmid are reported in (Antonovsky et al. 2016 [\[4\]](#)).

Growth tests

The growth test experiments were conducted in 96 well-plates. The final volume of each well was 200 μ L (50 μ L of mineral oil and 150 μ L culture). The media consisted of M9 media supplemented with varying concentrations of the relevant carbon source, and trace elements (without addition of vitamin B1). Bacterial cells were seeded from a culture tube. Growth temperature was set to 37°C, and either aerated with ambient air or air with elevated CO₂ (10%) either with ambient or reduced oxygen (5%). OD600 measurements were taken every 10-30 minutes using a Tecan Spark plate reader.

Chemostat evolution experiments

The evolutionary experiment was conducted in a Bioflo 110 chemostat (New Brunswick Scientific, USA) at a working volume of 0.7L and a dilution rate of 0.02h⁻¹ (equivalent to a doubling time of $\approx$ 33 hours) at 37°C. The chemostat was fed with media containing 4 g/L sodium formate and 0.5 g/L D-xylose as sole carbon sources. This amount of xylose in the feed makes xylose the limiting nutrient for cell growth in the chemostat. Gradually the concentration of xylose in the feed was reduced until a phenotype was observed. Chloramphenicol (30 mg/L) and streptomycin (100 mg/L) were added to the feed media. Aeration of the chemostat was done through a DASGIP MX4/4 stand-alone gas-mixing module (Eppendorf, Germany) with a composition of 10% CO₂ and 5% oxygen and 85% air at a flow rate of 40 sL/hr. To monitor the chemostat, a weekly sampling protocol was performed. Samples were taken for media analysis and phenotyping (inoculation of the bacteria on minimal media containing formate and lacking D-xylose). The optical density of each extracted sample was measured using a spectrophotometer (Ultrospec 10 Cell density meter, Amersham Biosciences) and a standard 10 mm polystyrene cuvette (Sarstedt, Germany).

¹³C Isotopic labelling experiment

A culture of cells that were growing on naturally labelled sodium formate in an elevated CO₂ (10%, naturally labelled) incubator (New Brunswick S41i CO₂ incubator shaker, Eppendorf, Germany) were diluted 10-fold into fresh M9 media with 30 mM ¹³C-formate sodium salt (Sigma Aldrich) to a total volume of 10 ml of culture. In the “open” labelling setup, growth was carried out in 125 ml glass shake flasks (n=3), which allow free exchange of gases between the headspace of the growth vessel and the gas mixture of the incubator. The flasks were placed inside an elevated CO₂ (10%) shaker-incubator (New Brunswick) with 37°C. After $\approx$ 4 doublings, the cells were

opening it. After ≈ 3 doublings, the cells were diluted 8-fold into flasks covered with air-tight rubber septa (SubaSeal, Sigma Aldrich). Then, the headspace of the flask was flushed with a gas mixture containing 10% $^{13}\text{CO}_2$ (Cambridge Isotope Laboratories, USA) + 90% air or 10% $^{12}\text{CO}_2$ + 90% air generated by a DASGIP MX4/4 stand-alone gas-mixing module (Eppendorf, Germany). The flasks were placed in a 37°C shaker incubator. After ≈ 4 doublings, the cells were harvested for subsequent analysis of protein bound amino acids. Glass flasks used in the labelling experiments were pretreated by heating in a 460°C furnace for 5 hours to evaporate any excess carbon sources that could remain in the vessels from previous utilizations. Each labelling experiment was conducted in triplicates ($n=3$), with $^{12}\text{CO}_2$ the triplicates were pooled and measured as three technical replicates. The predicted range of labelling under $^{13}\text{CO}_2$ was calculated using the following equation: $f_n = 1 - 0.5^n$, where f is the ^{13}C fraction and n is the number of doublings, as the remaining unlabelled biomass is decreased 2-fold at each doubling. The ^{13}C fraction was calculated for 3 and 5 doubling to create the predicted range.

Growth comparison between engineered autotrophic strains

The growth was conducted in a DASBox mini fermentation system (Eppendorf, Germany). The starting volume of each bioreactor was 150 ml M9 media supplemented with 60 mM Formate. The minimal media also included trace elements and vitamin B1 was omitted. The atmosphere was elevated 10% CO_2 5% O_2 and 85% N_2 . The culture was grown to saturation and then diluted in a ratio of 1:30, to obtain technical replicates.

Sample preparation for liquid chromatography coupled to mass spectrometry and mass analysis of biomass components

After harvesting the biomass culture samples were prepared and analyzed as described in (Antonovsky et al. 2016 [\[1\]](#)). Briefly, for protein bound amino acids ≈ 3 mL of culture at OD600 turbidity of ≈ 0.1 -0.15 were pelleted by centrifugation for 5 minutes at 8,000 g. The pellet was suspended in 1 mL of 6N HCl and incubated for 16 hours at 110°C. The acid was subsequently evaporated with a nitrogen stream, resulting in a dry hydrolysate. Dry hydrolysates were resuspended in 0.6 mL of miliQ water, centrifuged for 5 minutes at 14,000 g and the supernatant was used for injection into the LCMS. Hydrolyzed amino acids were separated using ultra performance liquid chromatography (UPLC, Acquity - Waters, USA) on a C-8 column (Zorbax Eclipse XBD - Agilent, USA) at a flow rate of 0.6 mL/min and eluted off the column using a hydrophobicity gradient. Buffers used were: A) H_2O + 0.1% formic acid and B) acetonitrile + 0.1% formic acid with the following gradient: 100% of A (0-3 min), 100% A to 100% B (3-9 min), 100% B (9-13 min), 100% B to 100% A (13-14 min), 100% A (14-20 min). The UPLC was coupled online to a triple quadrupole mass spectrometer (TQS - Waters, USA). Data was acquired using MassLynx v4.1 (Waters, USA). We selected amino acids which have peaks at distinct retention time and m/z values for all isotopologues and also showed correct ^{13}C labeling fractions in control samples that contained protein hydrolyzates of WT cells grown with known ratios of $^{13}\text{C}_6$ -glucose to ^{12}C -glucose.

The ^{13}C fraction of each metabolite was determined as the weighted average of the fractions of all the isotopologues of the metabolite, as depicted in the equation below:

$$\%^{13}\text{C} = \frac{\sum_{i=0}^n f_i \cdot i}{n}$$

where n is the number of carbons in the compound (e.g., for the amino acid serine $n=3$) and f_i is the relative fraction of the i -th isotopologue.

Whole-genome sequencing

DNA extraction (DNeasy blood & tissue kit, QIAGEN) and library preparation procedures were carried out as previously described in (Herz et al. 2017 [↗](#); Antonovsky et al. 2016 [↗](#)). The prepared libraries were sequenced by a Miseq machine (Illumina). Analysis of the sequencing data was performed as previously described in (Herz et al. 2017 [↗](#); Antonovsky et al. 2016 [↗](#); Gleizer et al. 2019 [↗](#)) with the breseq software (Barrick et al. 2014 [↗](#)). Cultivation Conditions for Metabolome Sampling

Cultivation Conditions for Metabolome Sampling

For the G6P and F6P, P5P, and S7P measurements: Single colonies were transferred into liquid 3 mL M9 media containing 0.2 g/L xylose, 2g/L formate, Chloramphenicol (30 mg/L) and streptomycin (100 mg/L) from fresh plates with the same medium. The M9 pre-cultures were adjusted to a starting OD600 of 0.05 into 10mL culture in 250 mL glass shake flasks. The flasks were placed inside an elevated CO₂ (10%) shaker-incubator (New Brunswick S41i CO₂ incubator shaker, Eppendorf, Germany) at 37°C. NADH/NADH⁺ measurements: Single colonies were transferred into 10 mL M9* xylose streptomycin from fresh M9* xylose streptomycin plates, and cultivated for 24h while shaking at 37°C. M9 pre-cultures were adjusted to a starting OD600 of 0.1 into 12-well plates, with 2 mL of medium in each well. Strains were cultivated in triplicates with or without formate (30 mM), added at the beginning of the culture. Optical density at 600nm was monitored every 10 min using a plate reader (Tecan, Spark). Plates were then rapidly transferred to a thermostatically controlled hood at 37°C and kept shaking during the sampling procedure.

Metabolomics Measurements

Cultivations were performed as described above. Culture aliquots were vacuum-filtered on a 0.45 µm pore size filter (HVLP02500, Merck Millipore). Filters were immediately transferred into a 40:40:20 (v-%) acetonitrile/methanol/water extraction solution at -20°C. Filters were incubated in the extraction solution for at least 30 minutes. Subsequently, metabolite extracts were centrifuged for 15 minutes at 13,000 rpm at -9°C and the supernatant was stored at -80°C until analysis. For measurements of NADH, NAD, sedoheptulose-7P and the pool of pentose phosphates, extracts were mixed with a ¹³C-labelled internal standard in a 1:1 ratio. LC-MS/MS analysis was performed with an Agilent 6495 triple quadrupole mass spectrometer (Agilent Technologies) as described previously (Donati et al. 2021 [↗](#)). An Agilent 1290 Infinity II UHPLC system (Agilent Technologies) was used for liquid chromatography. Temperature of the column oven was 30°C, and the injection volume was 3 µL. LC solvents in channel A were either water with 10 mM ammonium formate and 0.1% formic acid (v/v) (for acidic conditions, NADH/NAD), or water with 10 mM ammonium carbonate and 0.2% ammonium hydroxide (for basic conditions, sedoheptulose-7P and pool of pentoses). LC solvents in channel B were either acetonitrile with 0.1% formic acid (v/v) (for acidic conditions) or acetonitrile without additive (for basic conditions). LC columns were an Acquity BEH Amide (30 x 2.1 mm, 1.7 µm) for acidic conditions, and an iHILIC-Fusion(P) (50 x 2.1 mm, 5 µm) for basic conditions. The gradient for basic and acidic conditions was: 0 min 90% B; 1.3 min 40 % B; 1.5 min 40 % B; 1.7 min 90 % B; 2 min 90 % B. The ratio of ¹²C and ¹³C peak heights was used to quantify metabolites. ¹²C/¹³C ratios were normalized to OD at the time point of sampling. For the measurements of intracellular glucose-6-P and fructose-6-P, metabolic extracts were 10x concentrated using a vacuum evaporation (Eppendorf Concentrator plus) and resuspended in 40:40:20 (v-%) acetonitrile/methanol/water extraction solution. Temperature of the column oven was 30°C, and the injection volume was 5 µL. LC solvent in channel A was water with 10 mM ammonium formate and 0.1% formic acid (v/v), and in channel B was acetonitrile with 0.1% formic acid (v/v). The LC column was a Shodex HILICpak VG-50 2D (2.0 x 150 mm). The gradient was: 0 min 90% B; 58 min 70 % B; 3 min 90 % B. The ¹²C peak heights were used to quantify metabolites. Retention time of hexose-phosphates were determined with authentic standards of glucose-6-P

Chemostat proteomics experiment

The growth was conducted in a DASBox mini fermentation system (Eppendorf, Germany). The starting volume of each bioreactor was 150 ml M9 media supplemented with either 30 mM and 10mM D-xylose for the evolved and ancestor samples or only 10mM D-xylose for BW including compact set mutations. The minimal media also included trace elements and vitamin B1 was omitted. In experiments including the evolved strain and the ancestor strain, the atmosphere had elevated CO₂ (10%), and the chemostat dilution rate was set to 0.03 h⁻¹ corresponding to a doubling time of 24h.

Proteomics analysis

The cell pellets were lysed and subjected to in-solution tryptic digestion using the S-Trap method (by Protifi). The resulting peptides were analyzed using nanoflow liquid chromatography (nanoAcquity) coupled to high resolution, high mass accuracy mass spectrometry (Thermo Q Excative HFX). Each sample was analyzed on the instrument separately in discovery mode in a random order. Raw data was processed with MaxQuant v1.6.6.0. The data was searched with the Andromeda search engine against the E. coli BW25113 (K12) protein database as downloaded from Uniprot, appended with the mutant protein sequences and common lab protein contaminants. Search parameters included the following modifications: Fixed modification-cysteine carbamidomethylation. Variable modifications-methionine oxidation, and asparagine and glutamine deamidation. The quantitative comparisons were calculated using Perseus v1.6.0.7. Decoy hits were filtered out and only proteins that were detected in at least 2 replicates of at least one experimental group were kept.

Data, Materials, and Software Availability

All data and code are stored in a publicly available GitLab repository (<https://gitlab.com/milo-lab-public/compact-autocoli> .

Data S1. (GitLab)

iterative evolution mutations

Data S2. (GitLab)

Compact and evolved labeling run curated peaks summary

Data S3. (GitLab)

compact vs evolved growth experiment DATA 45mM 5 pctO₂

Data S4. (GitLab)

compact vs evolved growth experiment MAP 45mM 5 pctO₂

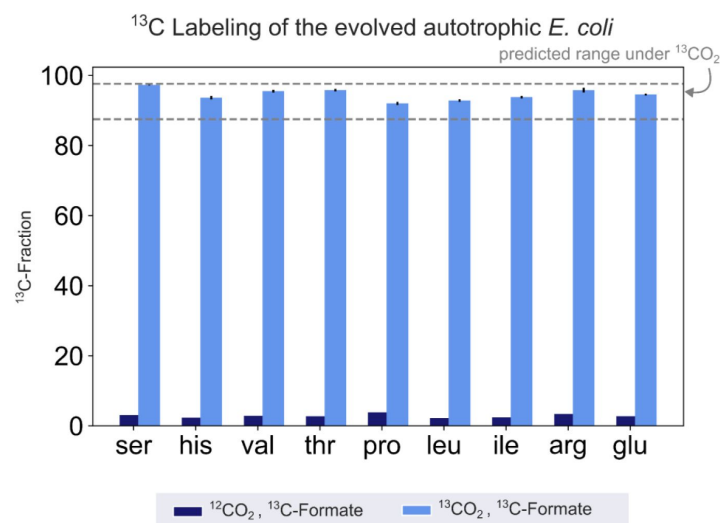


Fig. S1.

The fractional contribution of ¹³CO₂ to various protein-bound amino acids

evolved autotrophic strain after ≈4 doublings on ¹³CO₂ and ¹³C labelled formate (light blue, mean of n=3) reached the expected ¹³C labelling fraction of the biosynthesized amino acids (see methods). When grown in naturally labelled CO₂ and ¹³C labelled formate (dark blue, n=1, technical triplicate) the ¹³C content dropped to close to natural abundance. Experiments with ¹³CO₂ as the substrate were carried out in air-tight (i.e., sealed) growth vessels.

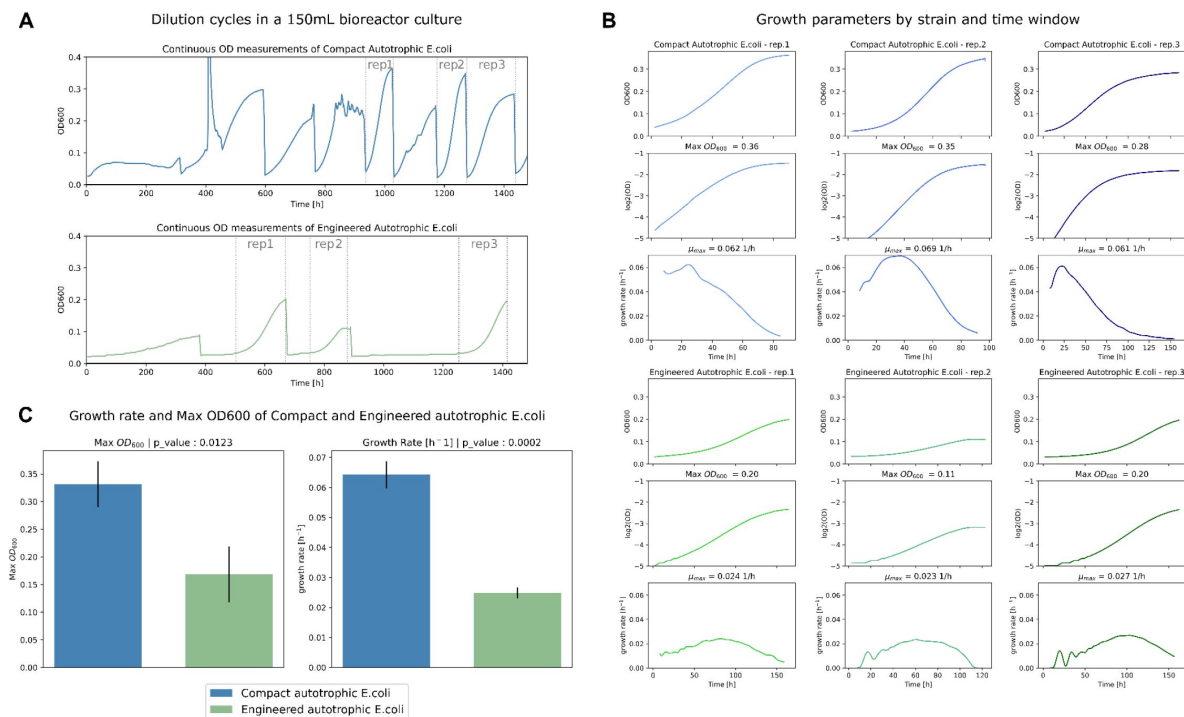


Fig. S2.

Growth of Compact versus Engineered autotrophic E. coli

(A) Growth curve of continuous cultures in liquid M9 minimal media supplemented with 60mM sodium formate and sparged with a gas mixture of 10% CO₂, 5% oxygen, and 85% nitrogen. Top: Compact autotrophic E. coli (green). Bottom: Engineered autotrophic strain (blue). The experiment was conducted in a DASBox mini fermentation system (Eppendorf, Germany) and the cultures were diluted in a 1:30 ratio once reaching stationarity. Dotted lines (grey) demarcate the manually selected time-windows used for the different technical replicates. **(B)** Growth parameters of the different technical replicates (as marked in panel A) that were used to calculate the maximal OD600. Growth rates were calculated by linear regression of the log-transformed OD600 values (data and source code can be found at <https://gitlab.com/milo-lab-public/compact-autocoli>). Top: replicates of the Compact autotrophic strain (green). Bottom: replicates of the Engineered autotrophic strain (blue). **(C)** Maximal OD600 and calculated growth rate of the Compact autotrophic E. coli (green) and the Engineered autotrophic E. coli (blue). Bar heights represent averages (n=3) of the maximal ODs and growth rates of each replicate. Error bars represent standard errors. P-values are based on Student's t-test with equal variance.

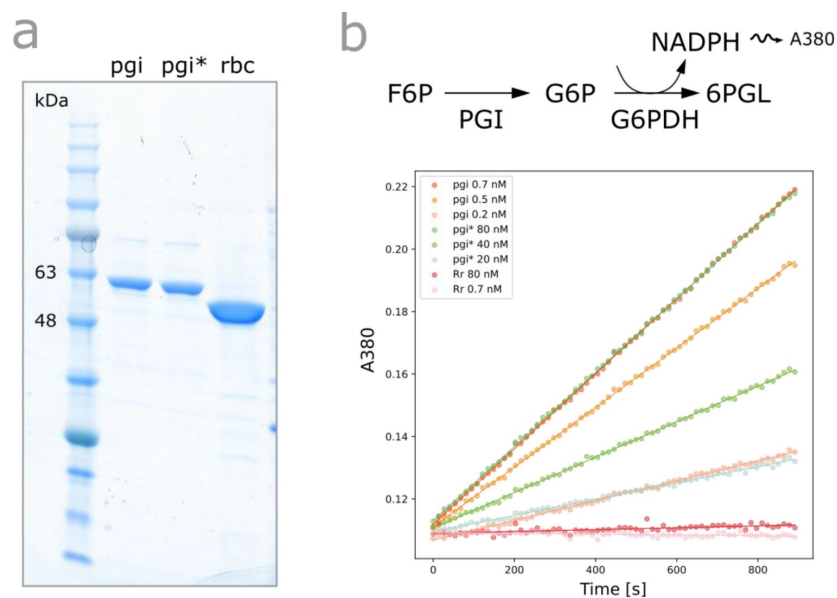


Fig. S3.

In vitro characterization of Pgi^{H386Y}.

(a) SDS-PAGE of purified Pgi wild-type, Pgi H386Y and Rubisco after expression in *E. coli*. **(b)** Spectrophotometric coupled assay determining the rate of the isomerization of fructose-6-phosphate (F6P) to glucose-6-phosphate (G6P) for purified Pgi wild-type and PgiH386Y. Rubisco was used as a negative control setting the noise level.

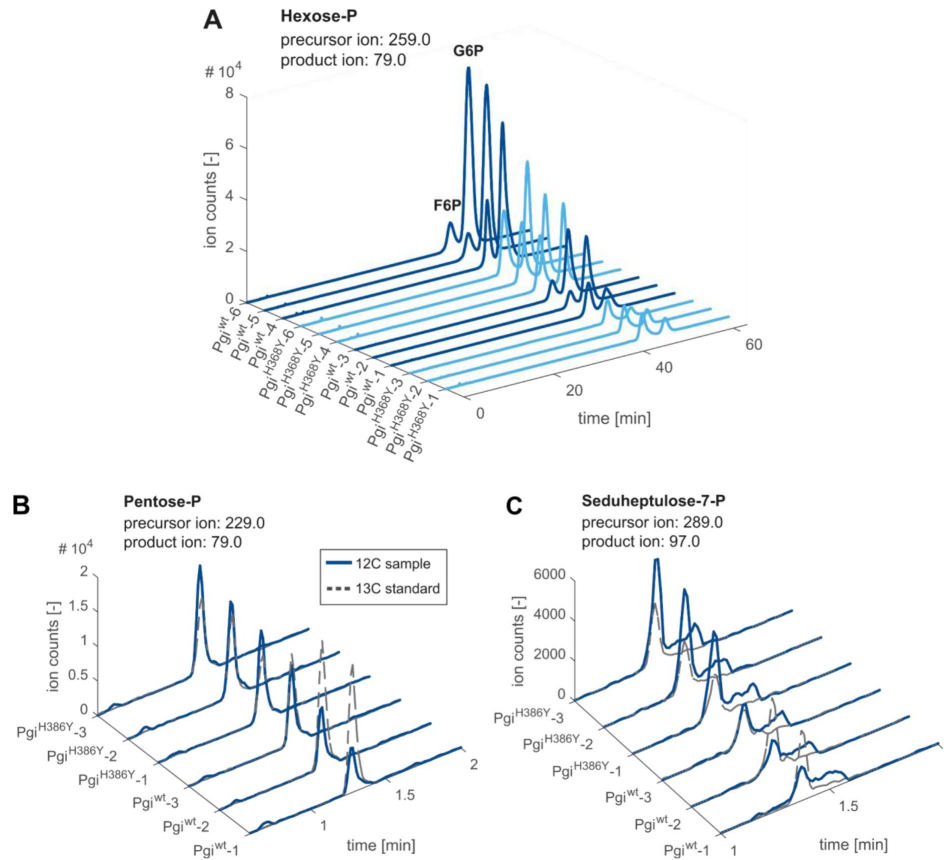


Fig. S4.

LC-MS/MS chromatogram of intracellular metabolites involved in the autocatalytic cycle in *E. coli* ancestor with Pgi^{H386Y} mutation and ancestor with a wild-type Pgi.

(A) Chromatogram of fructose-6-phosphate (F6P) and glucose-6-phosphate (G6P). Precursor ion (259.0) and product ion (79.0) were used for ¹²C detection. Strains (n=6) are grown in rubisco dependent conditions. Samples were measured using a Shodex HILICpak VG-50 2D. The retention time of hexose-phosphates were determined with authentic standards of G6P and F6P and the ¹²C peak heights were used to quantify metabolites. Experiments were performed in two batches of n=3 cultures. First batch samples (replicates 1-3) were used for measurements in A,B and C. **(B)** Chromatogram of total pool of pentose phosphates (ribulose-5P, ribose-5P, ribose-1P, and xylulose-5P - denoted P5P). Strains were grown in a rubisco-dependent manner (n=3) cultures. Overlay of ¹²C chromatograms (continuous line) and ¹³C chromatograms (dashed line) with precursor ion and product ion used for ¹²C detection as indicated. Samples were measured using an Acquity BEH Amide column and the ratio of ¹²C (sample) and ¹³C (internal standard) peak heights was used to quantify metabolites. **(C)** seduheptulose-7P (S7P) chromatogram of the same samples and conditions as indicated in B.

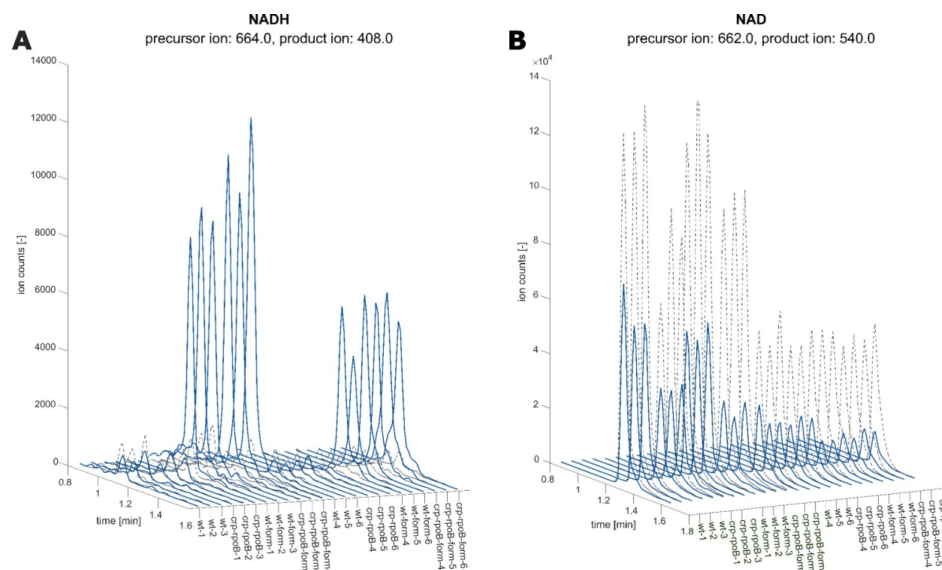


Fig. S5.

LC-MS/MS chromatograms of NADH and NAD in *fdh*-expressing *E. coli* in the presence and absence of formate.

(A) NADH chromatogram of *E. coli* BW25113 wild-type compared to a *crpH22N rpoBA1245V* mutant. Experiments were performed in two batches of $n=3$ cultures. Chromatograms are sorted accordingly. Overlay of ^{12}C chromatograms (continuous line) and ^{13}C chromatograms (dashed line). Precursor ion and product ion used for ^{12}C detection are indicated. Strains are grown with 30 mM formate (blue) or without the addition of formate (gray). **(B)** NAD chromatogram, same samples as in A. The ratio of ^{12}C (sample) and ^{13}C (internal standard) peak heights was used to quantify metabolites.

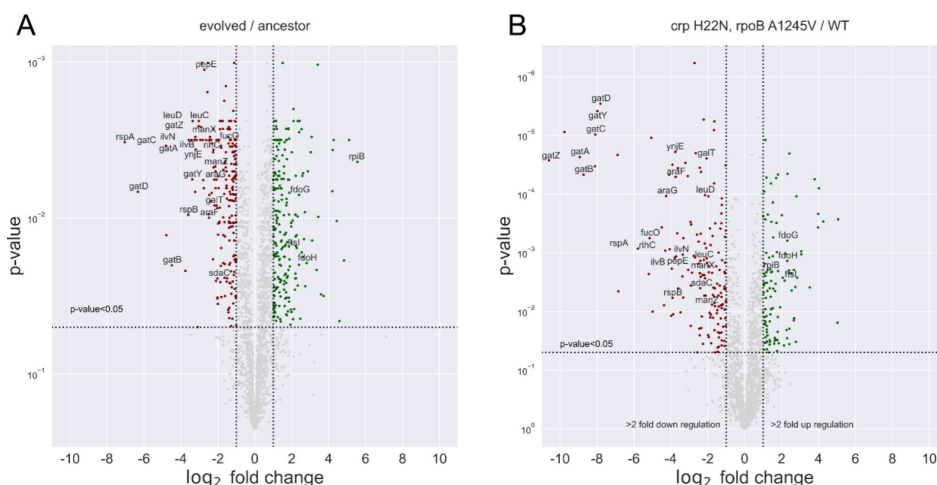


Fig. S6.

proteomics analysis of evolved strain and BW double-mutant

The differential protein expression between different strains. Red dots represent downregulated genes, Green - up-regulated. Only the genes that were had >4 fold change and <0.05 p-value in both sets were annotated. **(A)** Evolved / ancestor. **(B)** *rpoB* A1245V, *crp* H22N / WT.

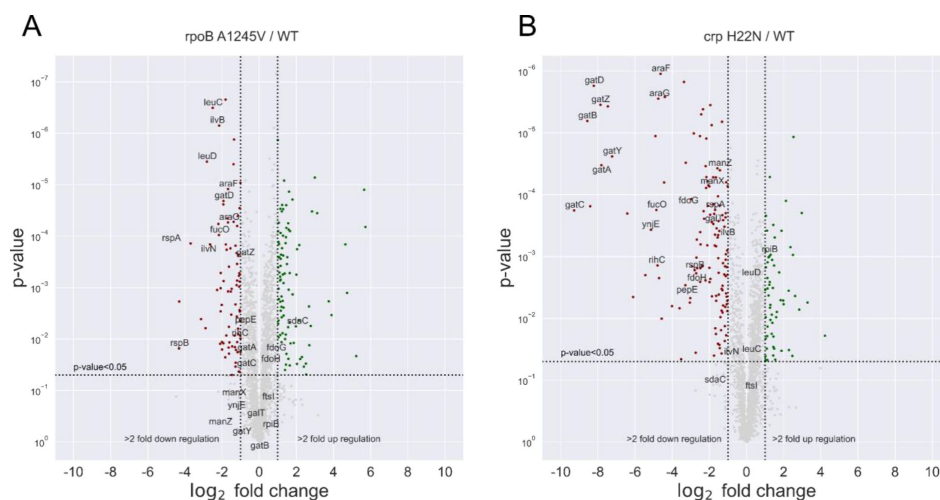


Fig. S7.

proteomics analysis of BW mutants

The differential protein expression between different strains. Red dots represent downregulated genes, Green - up-regulated. Only the genes that were had >4 fold change and <0.05 p-value in both of figure S6 A&B sets were annotated **(A)** *rpoB* A1245V / WT. **(B)** *crp* H22N / WT.

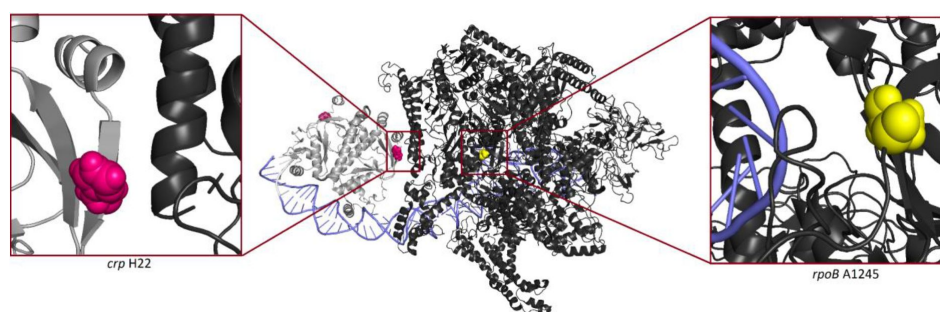


Fig. S8.

3D structure of wild type RNA polymerase and cAMP receptor protein

In light grey Crp, in black RNA polymerase (the holoenzyme which includes RpoB), in purple DNA, mutated positions are shown in pink (Crp H22) and in yellow (RpoB A1245). Crp H22 is located in activation region 2 of Crp. The protein structure was taken from [\(Shi et al. 2020\)](#).

Data S5. (GitLab)

pgi kinetic assays data

Data S6. (GitLab)

G6PvsF6PResults

Data S7. (GitLab)

PPP metabolites

Data S8. (GitLab)

Growth experiment WT vs DM FDH variants CO2 40mMFor DATA

Data S9. (GitLab)

Growth experiment WT vs DM FDH variants CO2 40mMFor MAP

Data S10. (GitLab)

Growth experiment WT vs DM FDH variants CO2 NoFormate DATA

Data S11. (GitLab)

Growth experiment WT vs DM FDH variants CO2 NoFormate MAP

Data S12. (GitLab)

rpoB crp metabolomics

Data S13. (GitLab)

proteomics Evo Anc

Data S14. (GitLab)

proteomics WT variants

Data S15. (GitLab)

Compact AutoColi dasgib log

Data S16. (GitLab)

Scaffoldless Autocoli dasgib log

Data S17. (GitLab)

CRP and RNA polymerase with mutations

References

1. Aidelberg Guy, Towbin Benjamin D., Rothschild Daphna, Dekel Erez, Bren Anat, Alon Uri (2014) **Hierarchy of Non-Glucose Sugars in Escherichia Coli** *BMC Systems Biology* **8**
2. Aigner H., Wilson R. H., Bracher A., Calisse L., Bhat J. Y., Hartl F. U., Hayer-Hartl M. (2017) **Plant RuBisCo Assembly in with Five Chloroplast Chaperones Including BSD2** *Science* **358**:1272–78
3. Ammar Ehab M., Wang Xiaoyi, Rao Christopher V. (2018) **Regulation of Metabolism in Escherichia Coli during Growth on Mixtures of the Non-Glucose Sugars: Arabinose, Lactose, and Xylose** *Scientific Reports* **8**
4. Antonovsky Niv, Gleizer Shmuel, Noor Elad, Zohar Yehudit, Herz Elad, Barenholz Uri, Zelcbuch Lior, et al. (2016) **Sugar Synthesis from CO₂ in Escherichia Coli** *Cell* **166**:115–25
5. Baba Tomoya, Ara Takeshi, Hasegawa Miki, Takai Yuki, Okumura Yoshiko, Baba Miki, Datsenko Kirill A., Tomita Masaru, Wanner Barry L., Mori Hirotsada (2006) **Construction of Escherichia Coli K-12 in-Frame, Single-Gene Knockout Mutants: The Keio Collection** *Molecular Systems Biology* **2**
6. Bang Junho, Hwang Chang Hun, Ahn Jung Ho, Lee Jong An, Lee Sang Yup (2020) **Escherichia Coli Is Engineered to Grow on CO₂ and Formic Acid** *Nature Microbiology* <https://doi.org/10.1038/s41564-020-00793-9>
7. Barenholz Uri, Davidi Dan, Reznik Ed, Bar-On Yinon, Antonovsky Niv, Noor Elad, Milo Ron (2017) **Design Principles of Autocatalytic Cycles Constrain Enzyme Kinetics and Force Low Substrate Saturation at Flux Branch Points** *eLife* **6** <https://doi.org/10.7554/eLife.20667>
8. Barrick Jeffrey E., Colburn Geoffrey, Deatherage Daniel E., Traverse Charles C., Strand Matthew D., Borges Jordan J., Knoester David B., Reba Aaron, Meyer Austin G. (2014) **Identifying Structural Variation in Haploid Microbial Genomes from Short-Read Resequencing Data Using Breseq** *BMC Genomics* **15**
9. Baumschabl Michael, Ata Özge, Mitic Bernd M., Lutz Lisa, Gassler Thomas, Troyer Christina, Hann Stephan, Mattanovich Diethard (2022) **Conversion of CO₂ into Organic Acids by Engineered Autotrophic Yeast** *Proceedings of the National Academy of Sciences of the United States of America* **119**
10. Charusanti Pep, Conrad Tom M., Knight Eric M., Venkataraman Karthik, Fong Nicole L., Xie Bin, Gao Yuan, Palsson Bernhard Ø. (2010) **Genetic Basis of Growth Adaptation of Escherichia Coli after Deletion of Pgi, a Major Metabolic Gene** *PLoS Genetics* **6**
11. Chen Frederic Y-H, Jung Hsin-Wei, Tsuei Chao-Yin, Liao James C. (2020) **Converting Escherichia Coli to a Synthetic Methyloph growing Solely on Methanol** *Cell* **182**:933–46
12. Cheng Kian-Kai, Lee Baek-Seok, Masuda Takeshi, Ito Takuro, Ikeda Kazutaka, Hirayama Akiyoshi, Deng Lingli, et al. (2014) **Global Metabolic Network Reorganization by Adaptive Mutations Allows Fast Growth of Escherichia Coli on Glycerol** *Nature Communications* **5**

13. Cherepanov P. P., Wackernagel W. (1995) **Gene Disruption in Escherichia Coli: TcR and KmR Cassettes with the Option of FLP-Catalyzed Excision of the Antibiotic-Resistance Determinant** *Gene* **158**:9–14
14. Claassens Nico J., Cotton Charles A. R., Kopljär Dennis, Bar-Even Arren (2019) **Making Quantitative Sense of Electromicrobial Production** *Nature Catalysis* <https://doi.org/10.1038/s41929-019-0272-0>
15. Conrad Tom M., Frazier Michael, Joyce Andrew R., Cho Byung-Kwan, Knight Eric M., Lewis Nathan E., Landick Robert, Palsson Bernhard Ø. (2010) **RNA Polymerase Mutants Found through Adaptive Evolution Reprogram Escherichia Coli for Optimal Growth in Minimal Media** *Proceedings of the National Academy of Sciences of the United States of America* **107**:20500–505
16. Davidi Dan, Noor Elad, Liebermeister Wolfram, Bar-Even Arren, Flamholz Avi, Tummeler Katja, Barenholz Uri, Goldenfeld Miki, Shlomi Tomer, Milo Ron (2016) **Global Characterization of in Vivo Enzyme Catalytic Rates and Their Correspondence to in Vitro Kcat Measurements** *Proceedings of the National Academy of Sciences of the United States of America* **113**:3401–6
17. Donati Stefano, Kuntz Michelle, Pahl Vanessa, Farke Niklas, Beuter Dominik, Glatter Timo, Gomes-Filho José Vicente, Randau Lennart, Wang Chun-Ying, Link Hannes (2021) **Multi-Omics Analysis of CRISPRi-Knockdowns Identifies Mechanisms That Buffer Decreases of Enzymes in E. Coli Metabolism** *Cell Systems* **12**:56–67
18. (2016) **“Engineering Cellular Metabolism.”** *Cell* **164**:1185–97
19. Flamholz Avi I., Dugan Eli, Blikstad Cecilia, Gleizer Shmuel, Ben-Nissan Roee, Amram Shira, Antonovsky Niv, et al. (2020) **Functional Reconstitution of a Bacterial CO₂ Concentrating Mechanism in Escherichia Coli** *eLife* **9** <https://doi.org/10.7554/eLife.59882>
20. Frendorf Pernille Ott, Lauritsen Ida, Sekowska Agnieszka, Danchin Antoine, Nørholm Morten H. H. (2019) **Mutations in the Global Transcription Factor CRP/CAP: Insights from Experimental Evolution and Deep Sequencing** *Computational and Structural Biotechnology Journal* **17**:730–36
21. Friedlingstein Pierre, O’Sullivan Michael, Jones Matthew W., Andrew Robbie M., Gregor Luke, Hauck Judith, Le Quéré Corinne, et al. (2022) **Global Carbon Budget 2022** *Earth System Science Data* **14**:4811–4900
22. Friedrich C. G., Bowien B., Friedrich B. (1979) **Formate and Oxalate Metabolism in Alcaligenes Eutrophus** *Journal of General Microbiology* **115**:185–92
23. Gassler Thomas, Sauer Michael, Gasser Brigitte, Egermeier Michael, Troyer Christina, Causon Tim, Hann Stephan, Mattanovich Diethard, Steiger Matthias G. (2020) **The Industrial Yeast Pichia Pastoris Is Converted from a Heterotroph into an Autotroph Capable of Growth on CO₂** *Nature Biotechnology* **38**:210–16
24. Gleizer Shmuel, Bar-On Yinon M., Ben-Nissan Roee, Milo Ron (2020) **Engineering Microbes to Produce Fuel, Commodities, and Food from CO₂** *Cell Reports Physical Science* **1**
25. Gleizer Shmuel, Ben-Nissan Roee, Bar-On Yinon M., Antonovsky Niv, Noor Elad, Zohar Yehudit, Jona Ghil, et al. (2019) **Conversion of Escherichia Coli to Generate All Biomass Carbon from CO₂** *Cell* **179**:1255–63

26. Grenier Frédéric, Matteau Dominick, Baby Vincent, Rodrigue Sébastien (2014) **Complete Genome Sequence of Escherichia Coli BW25113** *Genome Announcements* **2** <https://doi.org/10.1128/genomeA.01038-14>
27. van Rijsewijk Haverkorn, Bart R. B., Nanchen Annik, Nallet Sophie, Kleijn Roelco J., Sauer Uwe (2011) **Large-Scale 13C-Flux Analysis Reveals Distinct Transcriptional Control of Respiratory and Fermentative Metabolism in Escherichia Coli** *Molecular Systems Biology* **7**
28. Herz Elad, Antonovsky Niv, Bar-On Yinon, Davidi Dan, Gleizer Shmuel, Prywes Noam, Noda-Garcia Lianet, et al. (2017) **The Genetic Basis for the Adaptation of E. Coli to Sugar Synthesis from CO₂** *Nature Communications* **8**:1–10
29. Jackson Debra W., Simecka Jerry W., Romeo Tony (2002) **Catabolite Repression of Escherichia Coli Biofilm Formation** *Journal of Bacteriology* **184**:3406–10
30. Janasch Markus, Asplund-Samuelsson Johannes, Steuer Ralf, Hudson Elton P. (2019) **Kinetic Modeling of the Calvin Cycle Identifies Flux Control and Stable Metabolomes in Synechocystis Carbon Fixation** *Journal of Experimental Botany* **70**:973–83
31. Keller Philipp, Reiter Michael A., Kiefer Patrick, Gassler Thomas, Hemmerle Lucas, Christen Philipp, Noor Elad, Vorholt Julia A. (2022) **Generation of an Escherichia Coli Strain Growing on Methanol via the Ribulose Monophosphate Cycle** *Nature Communications* **13**
32. Kim Seohyoung, Lindner Steffen N., Aslan Selçuk, Yishai Oren, Wenk Sebastian, Schann Karin, Bar-Even Arren (2020) **Growth of E. Coli on Formate and Methanol via the Reductive Glycine Pathway** *Nature Chemical Biology* **16**:538–45
33. Landis L., Xu J., Johnson R. C. (1999) **The cAMP Receptor Protein CRP Can Function as an Osmoregulator of Transcription in Escherichia Coli** *Genes & Development* **13**:3081–91
34. Leger Dorian, Matassa Silvio, Noor Elad, Shepon Alon, Milo Ron, Bar-Even Arren (2021) **Photovoltaic-Driven Microbial Protein Production Can Use Land and Sunlight More Efficiently than Conventional Crops** *Proceedings of the National Academy of Sciences of the United States of America* **118** <https://doi.org/10.1073/pnas.2015025118>
35. Li Han, Opgenorth Paul H., Wernick David G., Rogers Steve, Wu Tung-Yun, Higashide Wendy, Malati Peter, Huo Yi-Xin, Cho Kwang Myung, Liao James C. (2012) **Integrated Electromicrobial Conversion of CO₂ to Higher Alcohols** *Science* **335**
36. Niu W., Kim Y., Tau G., Heyduk T., Ebright R. H. (1996) **Transcription Activation at Class II CAP-Dependent Promoters: Two Interactions between CAP and RNA Polymerase** *Cell* **87**:1123–34
37. Oh Jeong-Il, Bowien Botho (1998) **Structural Analysis of the Fds Operon Encoding the NAD⁺-Linked Formate Dehydrogenase of Ralstonia Eutropha *** *The Journal of Biological Chemistry* **273**:26349–60
38. Olofsson Jörgen, Hickler Thomas (2008) **Effects of Human Land-Use on the Global Carbon Cycle during the Last 6,000 Years** *Vegetation History and Archaeobotany* **17**:605–15
39. Phaneuf P. V., Gosting D., Palsson B. O. (2019) **"ALEdb 1.0: A Database of Mutations from Adaptive Laboratory Evolution Experimentation."** *Nucleic Acids*

40. Popov V. O., Lamzin V. S. (1994) **"NAD(+)-Dependent Formate Dehydrogenase."** *Biochemical Journal* **301**:625–43
41. Satanowski Ari, Dronsella Beau, Noor Elad, Vögeli Bastian, He Hai, Wichmann Philipp, Erb Tobias J., Lindner Steffen N., Bar-Even Arren (2020) **Awakening a Latent Carbon Fixation Cycle in Escherichia Coli** *Nature Communications* **11**
42. Savery N., Rhodius V., Busby S. (1996) **Protein-Protein Interactions during Transcription Activation: The Case of the Escherichia Coli Cyclic AMP Receptor Protein** *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* **351**:543–50
43. 42., Schwander Thomas, von Borzyskowski Lennart Schada, Burgener Simon, Cortina Niña Socorro, Erb Tobias J. (2016) **"A Synthetic Pathway for the Fixation of Carbon Dioxide in Vitro."** *Science* **354**:900–904
44. Shi Wei, Jiang Yanan, Deng Yibin, Dong Zigang, Liu Bin (2020) **Visualization of Two Architectures in Class-II CAP-Dependent Transcription Activation** *PLoS Biology* **18**
45. Tenaillon Olivier, Rodríguez-Verdugo Alejandra, Gaut Rebecca L., McDonald Pamela, Bennett Albert F., Long Anthony D., Gaut Brandon S. (2012) **The Molecular Diversity of Adaptive Convergence** *Science* **335**:457–61
46. Thomason Lynn C., Costantino Nina, Court Donald L. (2007) **"E. Coli Genome Manipulation by P1 Transduction."** *Current Protocols in Molecular Biology / Edited by Frederick M. Ausubel ... [et Al.]*
47. Utrilla Jose, O'Brien Edward J., Chen Ke, McCloskey Douglas, Cheung Jacky, Wang Harris, Armenta-Medina Dagoberto, Feist Adam M., Palsson Bernhard O. (2016) **Global Rebalancing of Cellular Resources by Pleiotropic Point Mutations Illustrates a Multi-Scale Mechanism of Adaptive Evolution** *Cell Systems* **2**:260–71
48. Van Peteghem L., Sakarika M., Matassa S., Pikaar I., Ganigué R., Rabaey K. (2022) **Towards New Carbon-Neutral Food Systems: Combining Carbon Capture and Utilization with Microbial Protein Production** *Bioresource Technology* **349**
49. Widjaja Arief, Shiroshima Masahiro, Oka Takashi, Yasuda Masahiro, Ogino Hiroyasu, Miyatake Kazutaka, Nakajima Hiroshi, Ishikawa Haruo (1998) **The Kinetics and Mechanism of a Reaction Catalyzed by Bacillus Stearothermophilus Phosphoglucose Isomerase** *Journal of Fermentation and Bioengineering* **86**:324–31
50. Yishai Oren, Bouzon Madeleine, Döring Volker, Bar-Even Arren (2018) **In Vivo Assimilation of One-Carbon via a Synthetic Reductive Glycine Pathway in Escherichia Coli** *ACS Synthetic Biology* **7**:2023–28
51. Zelcbuch Lior, Antonovsky Niv, Bar-Even Arren, Levin-Karp Ayelet, Barenholz Uri, Dayagi Michal, Liebermeister Wolfram, et al. (2013) **Spanning High-Dimensional Expression Space Using Ribosome-Binding Site Combinatorics** *Nucleic Acids Research* **41**

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Joint Public Review:

The authors previously showed that expressing formate dehydrogenase, rubisco, carbonic anhydrase, and phosphoribulokinase in *Escherichia coli*, followed by experimental evolution, led to the generation of strains that can metabolise CO₂. Using two rounds of experimental evolution, the authors identify mutations in three genes - *pgi*, *rpoB*, and *crp* - that allow cells to metabolise CO₂ in their engineered strain background. The authors make a strong case that mutations in *pgi* are loss-of-function mutations that prevent metabolic efflux from the reductive pentose phosphate autocatalytic cycle. The authors also use proteomic analysis to probe the role of the mutations in *crp* and *rpoB*. While they do not reach strong conclusions about how these mutations promote autotrophic growth, they provide some clues, leading to valuable speculation.

Comments on revised version:

The authors have thoroughly addressed the reviewers' comments. The major addition to the paper is the proteomic analysis of single and double mutants of *crp* and *rpoB*. These new data provide clues as to the role of the *crp* and *rpoB* mutations in promoting autotrophic growth, which the authors discuss. The authors acknowledge that it will require additional experiments to determine whether the speculated mechanisms are correct. Nonetheless, the new data provide valuable new insight into the role of the *crp* and *rpoB* mutations. The authors have also expanded their description of the *crp* and *rpoB* mutations, making it clearer that the effects of these mutations are likely to be distinct, albeit with potential for overlap in function.

<https://doi.org/10.7554/eLife.88793.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Author response:

Reviewer #1:

The main objective of this study is to achieve the development of a synthetic autotroph using adaptive laboratory evolution. To accomplish this, the authors conducted chemostat cultivation of engineered E. coli strains under xylose-limiting conditions and identified autotrophic growth and the causative mutations. Additionally, the mutational mechanisms underlying these causative mutations were also explored with drill down assays. Overall, the authors demonstrated that only a small number of genetic changes were sufficient (i.e., 3) to construct an autotrophic E. coli when additional heterologous genes were added. While natural autotrophic microorganisms typically exhibit low genetic tractability, numerous studies have focused on constructing synthetic autotrophs using platform microorganisms such as E. coli. Consequently, this research will be of interest to synthetic biologists and systems biologists working on the development of synthetic autotrophic microorganisms. The conclusions of this paper are mostly well supported by appropriate experimental methods and logical reasoning. However, further experimental validation of the mutational mechanisms involving rpoB and crp would enhance readers' understanding and provide clearer insights, despite acknowledgement that these genes impact a broad set of additional genes. Additionally, a similar study, 10.1371/journal.pgen.1001186, where pgi was deleted from the E. coli genome and evolved to reveal an rpoB mutation is relevant to this work and should be placed in the context of the presented findings.

We thank the reviewer for pointing this study out. It is very interesting that a mutation in a similar region in RpoB was observed in a related context of Pgi loss of activity. We have added a reference to this study in our text (Page 11, line 21).

he authors addressed rpoB and crp as one unit and performed validation. They cultivated the mutant strain and wild type in a minimal xylose medium with or without formate, comparing their growth and NADH levels. The authors argued that the increased NADH level in the mutant strain might facilitate autotrophic growth. Although these phenotypes appear to be closely related, their relationship cannot be definitively concluded based on the findings presented in this paper alone. Therefore, one recommendation is to explore investigating transcriptomic changes induced by the rpoB and crp mutations. Otherwise, conducting experimental verification to determine whether the NADH level directly causes autotrophic growth would provide further support for the authors' claim.

We appreciate the valuable comment and agree that the work was lacking such an analysis. Due to various reasons we have opted to use a proteomic approach which we feel fulfills the same purpose as the transcriptomics suggestion. We found interesting evidence in up-regulation of the fdoGH operon (comprising the native formate dehydrogenase O enzyme complex) which could indicate why there is an increase in NADH/NAD⁺ levels. We also hypothesize that this upregulation might be important more generally by drawing comparisons to natural chemo-autotrophs.

Further experimental work (which we were not able to include in the current study) could help validate this link by deleting fdoGH and observing a loss of phenotype and, on the flip side, directly overexpressing the fdoGH operon and observing an increase in the NADH/NAD⁺ ratio. Indeed, if this overexpression were to prove sufficient for achieving an autotrophic

phenotype without the mutations in the global transcription regulators, it would be a much more transparent design.

We have added a section titled "Proteomic analysis reveals up-regulation of rPP cycle and formate-associated genes alongside down-regulation of catabolic genes" to the Results based on this analysis.

- *It would be beneficial to provide a more detailed explanation of the genetic background before the evolution stage, specifically regarding the Δpfk and Δzwf mutations. Furthermore, it is suggested to include a figure that provides a comprehensive depiction of the reductive pentose phosphate pathway and the bypass pathway. These will help readers grasp the concept of the "metabolic scaffold" as proposed by the authors.*

We agree with the reviewer that this could be helpful and we added a reference to the original paper Gleizer et al. 2019 that reported this design and also includes the relevant figure. We feel that the figure should not be added to the current manuscript as we continue to show that this design is not relevant in the context of the three reported mutations and such a figure could distract the attention of the reader from the main takeaways of the current study.

- *Despite the essentiality of the $rpoB$ mutation (A1245V) to the autotrophic phenotype in the final strain, the inclusion of this mutation in step C1 does not appear to be justified. According to line 37 on page 3, the authors chose to retain the unintended mutation in $rpoB$ based on its essentiality to the phenotype observed in other evolved strains. However, it should be noted that the mutations found in the evolved strain I, II, and III (P552T or D866E) were entirely different from the unintended mutation (A1245V) during genetic engineering. This aspect should be revised to avoid confusion among readers.*

Thank you for pointing this issue out, we added a clarification in the text (page 4 line 7) to avoid such confusion. We believe this point is much clearer now.

*The $rpoB$ mutation which was shown to be essential in the study is indeed known to be common in ALE experiments in *E. coli*. Thus, I searched the different $rpoB$ mutations in ALEdb in *E. coli* and I was able to find a similar mutation in a study where pgi was knocked out and then evolved. <https://doi.org/10.1371/journal.pgen.1001186> This study seems very relevant given that pgi was a key mutation in the compact set of this work and the section "Modulation of a metabolic branch-point activity increased the concentration of rPP metabolites" informs that loss of function mutations in pgi were also found. The findings of this study should thus be put in the context of the previous related ALE study. I would recommend a similar analysis of crp mutations from studies in ALEdb to see if there are similar mutations in this gene as well or if this a unique mutation.*

We thank the reviewer for bringing this publication to our attention. We have addressed this observation in the main text (page 11, line 21). We agree that it could have some connection to the pgi mutation yet we would not want to overspeculate about this role, as we also found the exact same mutation (A1245V) as an adaptation to higher temperature in another *E. coli* study (Tenaillon et al. 2012). We would like to bring forward the fact that the two reported $rpoB$ mutations are always accompanied by another mutation with pleiotropic effects, either in the transcription factor Crp or in another RNA polymerase subunit (e.g. RpoC). As such many epistatic effects could occur, one of which we also report here in page 13, line 18. In conclusion, although there could be a connection between the $rpoB$ and pgi mutations, it

could be a mere coincidence and the two mutations could exhibit two distinct roles in two distinct phenotypes.

We also would like to thank the reviewer for suggesting a similar analysis for *crp* and found another mutation at a nearby residue with strong adaptive effects and mentioned it in our main text.

*Can the typical number of mutations found in a given ALE experiment be directly compared to those found in this study? It seems like a retrospective analysis of other ALE studies to show how many mutations typically occur in an ALE study and sets which were found to be causal to reproduce the phenotype of interest (through similar reverse engineering in the starting strain) should be presented. Again, the authors cite ALEdb which should provide direct numbers of mutations found in similar ALE studies with *E. coli* and one could then examine them to find sets of clearly causal mutations which recreate phenotypes of interest. Such an analysis would go a long way in supporting the main finding of "small number" of mutations.*

Discussion, page 12, line 42. "This could serve as a promising strategy for achieving minimally perturbed genotypes in future metabolic engineering attempts". There is an entire body of work around growth-coupled production which can be predicted and evolved with a genome-scale metabolic model and ALE. Thus, if this statement is going to be made, relevant studies should be cited and placed in context.

The reviewer raises an important point which could indeed yield an interesting perspective. However, it would be difficult to perform this comparison in practice since many of the studies published on ALEdb have not isolated essential mutations from other mutation incidents nor have they determined the role of each mutation in the reported phenotypes. For example, many ALE trajectories include a hypermutator that greatly increases the number of irrelevant mutations and it is nearly impossible to sieve through them to find an essential set.

Moreover, it is hard to compare the "level of difficulty" of achieving one phenotype over another and therefore feel that even though such an analysis would be insightful, it requires an amount of work which is outside the scope of this study.

Finally, we would like to highlight our approach of using the iterative approach, isolating the relevant consensus mutations and repeating this process until no evolution process is required, we are not aware of prior studies that used this approach.

We now clarified what we mean by "promising strategy" in the discussion in order to avoid any false claims about novelty (page 16 line 32): "Using metabolic growth-coupling as a temporary 'metabolic scaffold' that can be removed, could serve as a promising strategy for achieving minimally perturbed genotypes in future metabolic engineering attempts."

Reviewer #2:

*Synthetic autotrophy of biotechnologically relevant microorganisms offers exciting chances for CO₂ neutral or even CO₂ negative production of goods. The authors' lab has recently published an engineered and evolved *Escherichia coli* strain that can grow on CO₂ as its only carbon source. Lab evolution was necessary to achieve growth. Evolved strains displayed tens of mutations, of which likely not all are necessary for the desired phenotype.*

*In the present paper the authors identify the mutations that are necessary and sufficient to enable autotrophic growth of engineered *E. coli*. Three mutations were identified, and their phenotypic role in enhancing growth via the introduced Calvin-Benson-Bassham*

cycle were characterized. It was demonstrated that these mutations allow autotrophic growth of *E. coli* with the introduced CBB cycle without any further metabolic intervention. Autotrophic growth is demonstrated by ¹³C labelling with ¹³C CO₂, measured in proteinogenic amino acids. In Figures 2B and S1, the labeling data are shown, with an interval of the "predicted range under ¹³CO₂".

Here, the authors should describe how this interval was derived.

The methodology is clearly described and appropriate.

The present results will allow other labs to engineer *E. coli* and other microorganisms further to assimilate CO₂ efficiently into biomass and metabolic products. The importance is evident in the opportunity to employ such strain in CO₂ based biotech processes for the production of food and feed protein or chemicals, to reduce atmospheric CO₂ levels and the consumption of fossil resources.

Please describe in the methodology how the interval of the predicted range of ¹³C labeling was derived for Figures 2B and S1. Was it calculated by the dilution factor during 4 generations, or did you predict the label incorporation individually with a metabolic model?

The text needs careful editing, some sentences are incomplete and there are frequent inconsistencies in writing metabolites and enzymes.

P2L6: unclear sentence (incomplete?)

P2L19: *pastoris* with lower case "p"

P2L40: incomplete sentence

P2L42: here, and at many other places, the writing of RuBisCO needs to be aligned. It is an abbreviation and should begin with a capital letter. Most commonly it is written as RuBisCO which I would suggest - please unify throughout the text.

P3L3: formate dehydrogenase ... metabolites and enzymes with lower case letter. And, no hyphen here.

P5L4: delete the : after unintentionally

P6L16: carboxylation of RuBP (it is not CO₂ that is carboxylated - if any, CO₂ is carboxylating)

P7L25: phosphoglucosomerase (lower case)

P8L5: in line

P8L9: part of glycolysis/ ...

P10L4: pentose phosphates (lower case, no hyphen).

P10L4: all metabolites lower case

P12L28: incomplete sentence

P18L4: *Escherichia coli* in italics P18L15: *Pseudomonas sp.* in italics P18L16: ... promoter and with a strong ...

P20, chapter Metabolomics: put the numbers of ¹²C and ¹³C in superscript P23L9: pentose phosphates ; all metabolites in lower case (as above) P23: all ¹²C and ¹³C with superscript numbers.

We thank the reviewer for their comments, and for pointing out the need to clarify how we derived the predicted range of ^{13}C labeling. We edited the text accordingly, and added the relevant calculation to the methods section (under the “ ^{13}C Isotopic labeling experiment”). We would like to also thank the reviewer for the required text improvements, which were implemented.

Reviewer #3:

*The authors previously showed that expressing formate dehydrogenase, rubisco, carbonic anhydrase, and phosphoribulokinase in Escherichia coli, followed by experimental evolution, led to the generation of strains that can metabolise CO_2 . Using two rounds of experimental evolution, the authors identify mutations in three genes - *pgi*, *rpoB*, and *crp* - that allow cells to metabolise CO_2 in their engineered strain background. The authors make a strong case that mutations in *pgi* are loss-of-function mutations that prevent metabolic efflux from the reductive pentose phosphate autocatalytic cycle. The authors also argue that mutations in *crp* and *rpoB* lead to an increase in the NADH/NAD^+ ratio, which would increase the concentration of the electron donor for carbon fixation. While this may explain the role of the *crp* and *rpoB* mutations, there is good reason to think that the two mutations have independent effects, and that the change in NADH/NAD^+ ratio may not be the major reason for their importance in the CO_2 -metabolising strain.*

We thank the reviewer for their comments and constructive feedback.

We agree that there is probably a broader effect caused by the *rpoB* and *crp* mutations, besides the change in the NADH/NAD^+ ratio. Hence, we performed a proteomics analysis, comparing the *rpoB* and *crp* mutations on a WT background to an autotrophic *E. coli*, searching for a mutual change in both strains compared to their “ancestors”. We found up-regulation of rPP cycle and formate-associated genes, and a down-regulation of catabolic genes. We added a section dedicated to this matter under the title “Proteomic analysis reveals up-regulation of rPP cycle and formate-associated genes alongside down-regulation of catabolic genes”.

Specific comments:

1. Deleting *pgi* rather than using a point mutation would allow the authors to more rigorously test whether loss-of-function mutants are being selected for in their experimental evolution pipeline. The same argument applies to *crp*.

We appreciate this recommendation and indeed tried to delete *pgi*, but the genetic manipulation caused a knockout of other genes along with *pgi* (*pepE*, *rluF*, *yjbD*, *lysC*) so in the time available to us we cannot confidently determine whether the deletion alone is sufficient and can replace the mutation.

Regarding *crp*, we do not think there is a reason to believe the mutation is a loss-of-function. In any case, the proteomics-based characterization of the *crp* mutation is now included in the SI.

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